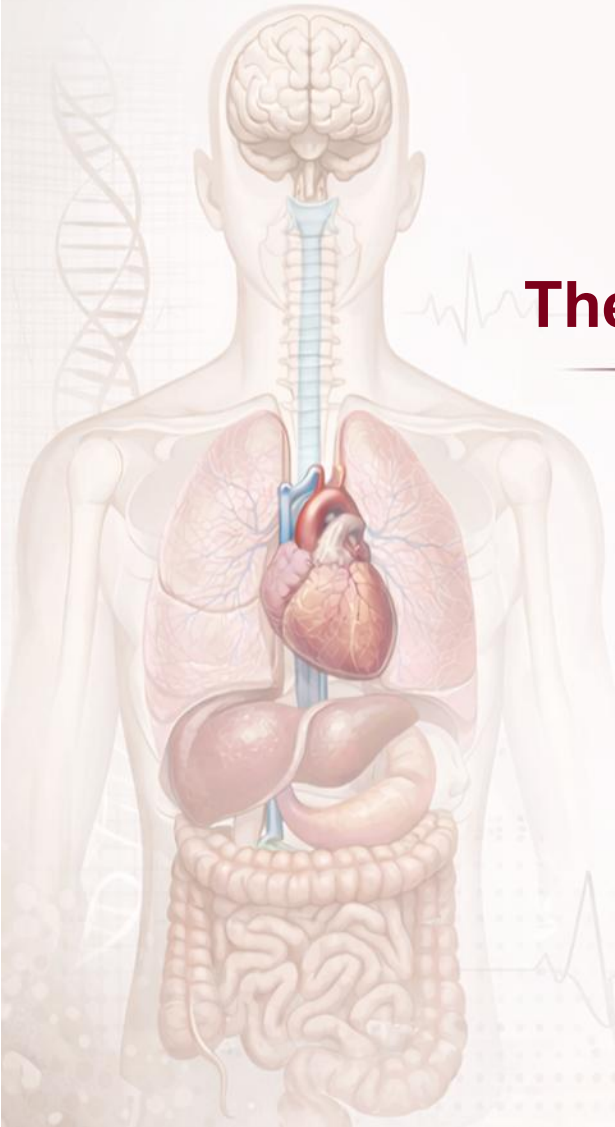
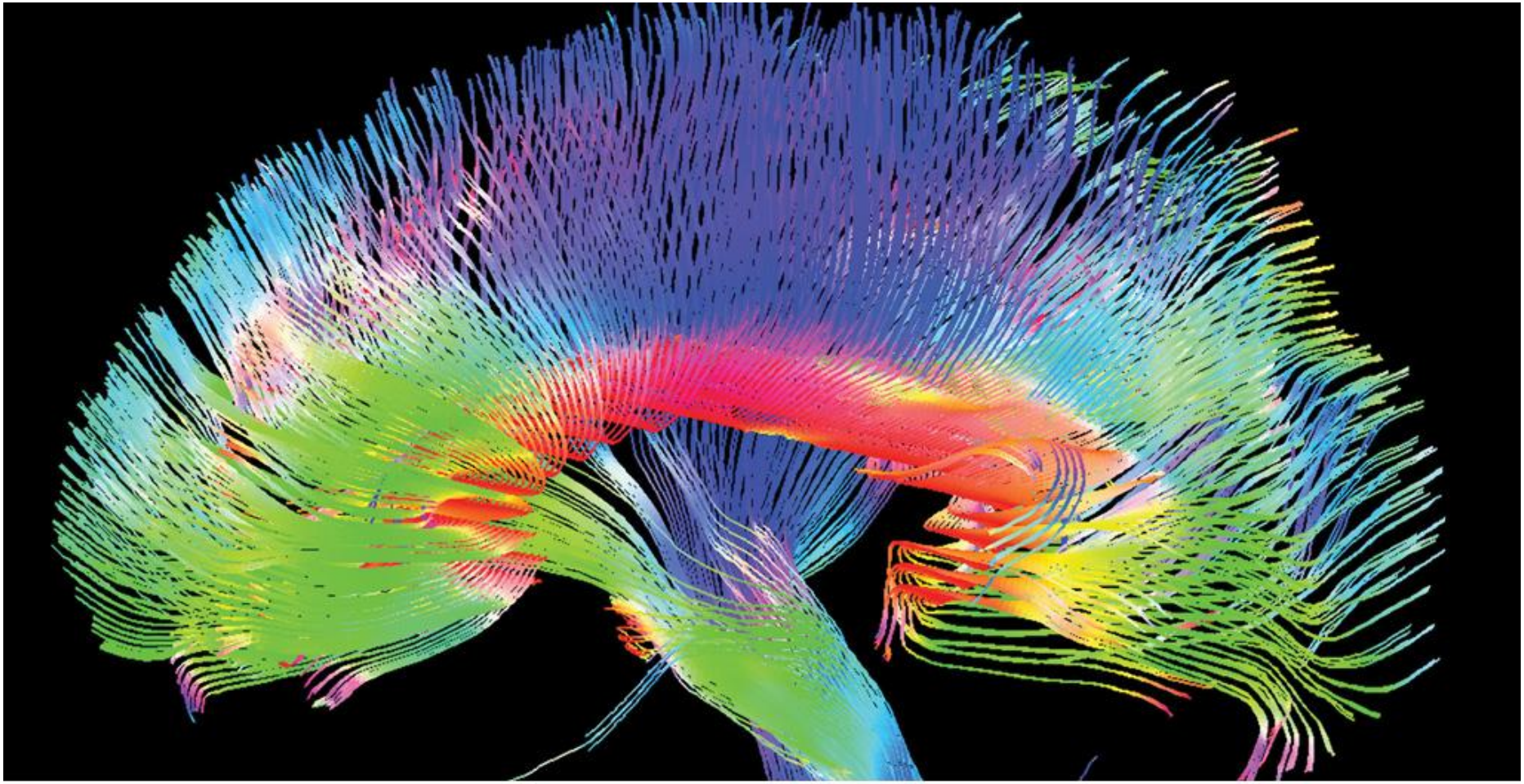


Chapter 5.
The Central Nervous System

KOREA UNIVERSITY
1905

Transmission electron micrograph of two synapses



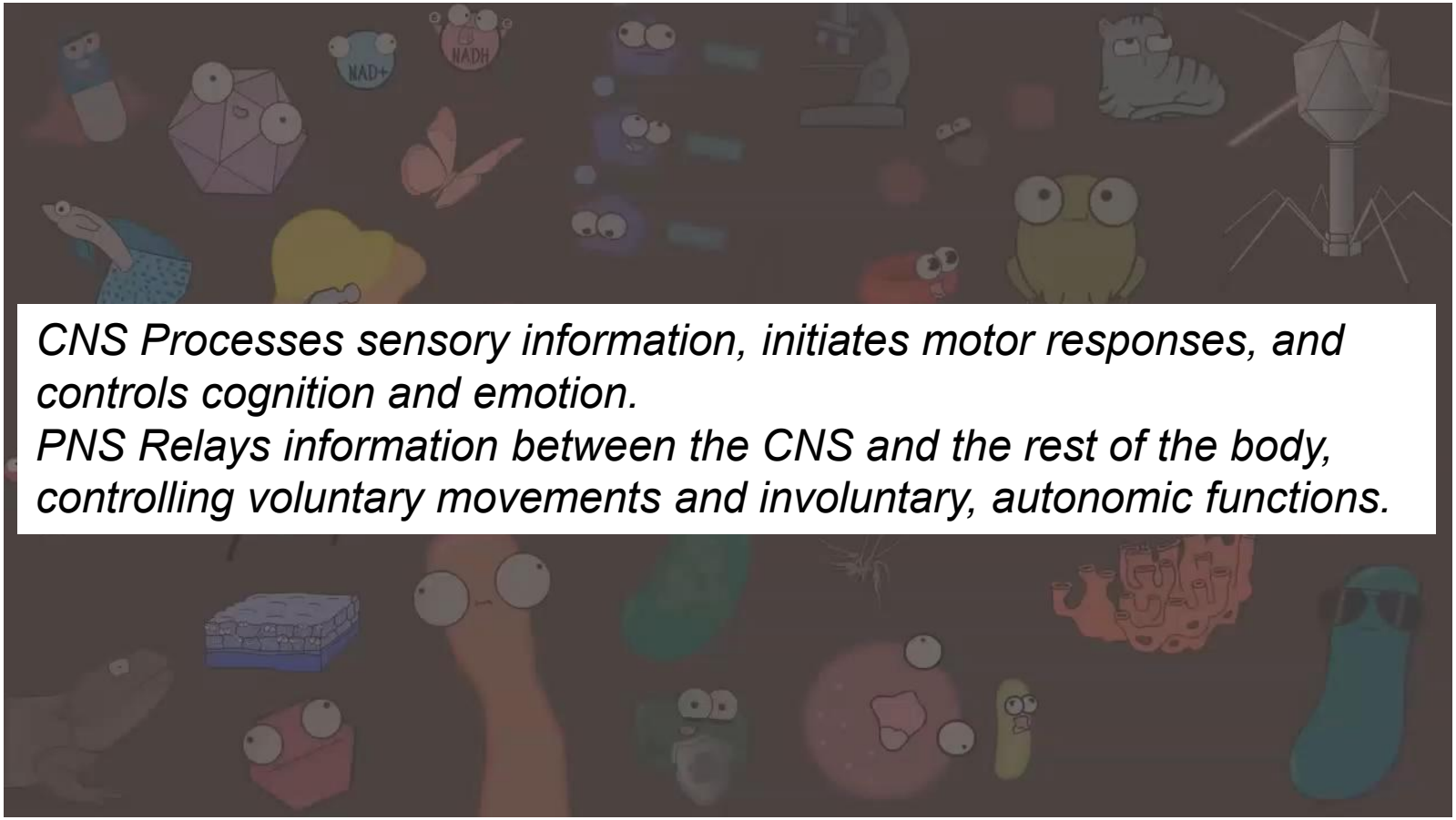
Tom Barrick,Chris Clark,Sghms/Science Source

Staining of white matters

White matter: myelinated nerve fibers, connecting different parts of brain

Blue: top-bottom, Green: front-back, Red: left-right halves of the brain

Nervous system



CNS Processes sensory information, initiates motor responses, and controls cognition and emotion.
PNS Relays information between the CNS and the rest of the body, controlling voluntary movements and involuntary, autonomic functions.

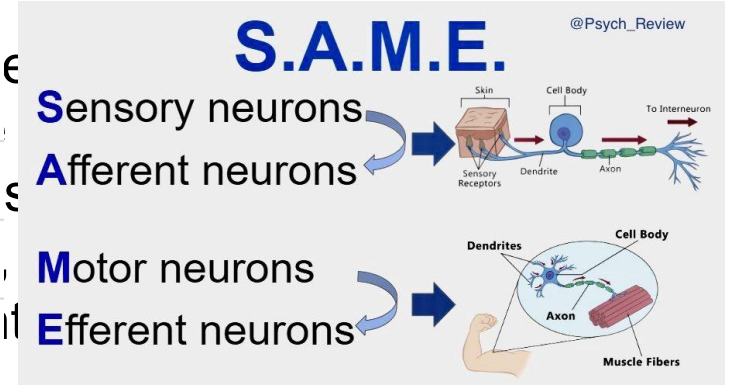
https://www.youtube.com/watch?v=RNLceVI8jcc

Central nervous system in terms of homeostasis

- The two major regulatory systems of the body that ensure life-sustaining coordinated responses are *nervous and endocrine systems*
- *Afferent, efferent, and interneurons* form a complex interactive network of excitable cells and 90% of nervous system cells are *nonexcitable* glial cells, which structurally and functionally interact with neurons

Key Differences at a Glance

Neuron Type	Also Known As	Function	Direction
Afferent	Sensory Neurons	Detect/transmit sensations	Towards CNS
Efferent	Motor Neurons	Actuate muscles/glands	Away from CNS
Interneuron	Association Neurons	Relay signals/processing	Within CNS



- These neuron-controlled activities are directed toward homeostasis
- Electrical signals (action potentials) and neurotransmitters released are to control the rapid responses of the body

5.1 Organization and Cells of the Nervous System

Use it or lose it

- During the development, nervous system develops according to the genetic plan, an overabundance of neurons and synapses is formed
- Depending on external stimuli and the usage of each neural pathway, some are retained while the others are eliminated
- Amblyopia (lazy eye; 약시): weaker eye is less used for vision
  - Functionally blind eye itself is normal
  - Defect lies in the lost neural connections in brains' visual pathway
  - If weaker eye is forced to work by covering the stronger eye during the developmental period, the weaker eye can retain full vision



- Maturation of the nervous system: "use it or lose it"
- Even after its maturation, the nervous system continues to learn

Organizations and cells of the nervous system

- The nervous system is organized into the central (CNS) and peripheral (PNS) nervous system
  - Central nervous system (CNS): brain and spinal cord (interneuron division, 연합/사이신경세포)
  - Peripheral nervous system (PNS): nerve fibers connecting CNS and other parts of the body (peripheral side)
  - PNS is further divided into afferent and efferent divisions

- In addition to CNS and PNS, the enteric nervous system (ENS) is an extensive nerve network in the wall of digestive system
  - Digestive systems are controlled by autonomic nervous system of PNS, ENS, and hormones

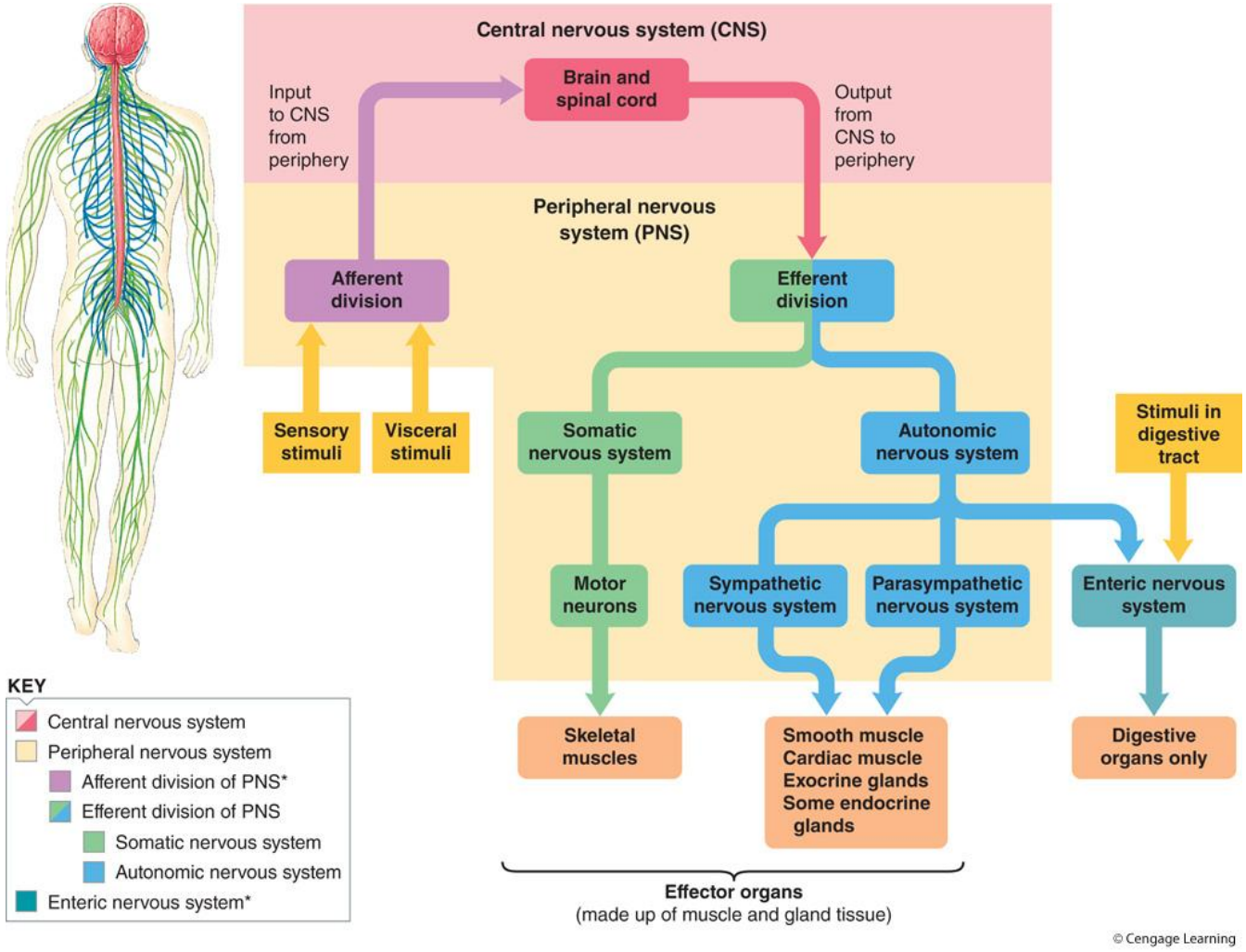
- All these nervous systems are subdivisions of a single integrated nervous system, different in the structure, location, and functions inside the whole nervous system

Two divisions of the PNS

- Afferent (=carrying toward, 감각/들신경세포) neurons
  - Carry information to CNS, including evaluation of the external environment and status report on the internal activities

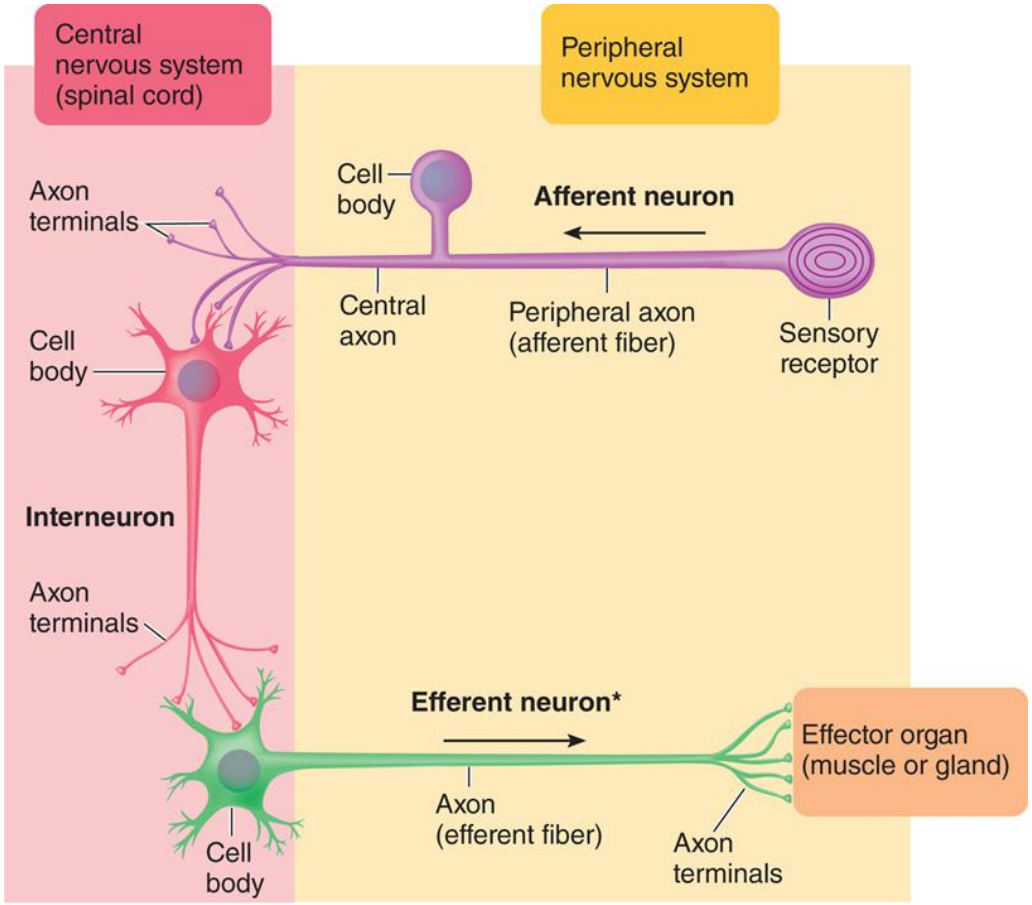
- Efferent (=carrying from, 운동/날신경세포) neurons
  - Instruction from CNS are transmitted via efferent division to effector organs
  - Further divided into somatic (SNS) and autonomic (ANS) nervous system
  - SNS: fibers of the motor neurons supporting the voluntary movements of skeletal muscles
  - ANS: fibers innervating smooth muscle, cardiac muscle, and glands regulating their involuntary functions
  - ANS is further divided into sympathetic and parasympathetic nervous system

Organizations and cells of the nervous system



Three functional classes of neurons

- Afferent neurons
  - sensory neurons that supply input to the CNS about the environment
- Interneurons
  - carry instructions from CNS to organs, muscles and glands
- Efferent neurons
  - responsible for integrating afferent information and formulating an efferent response



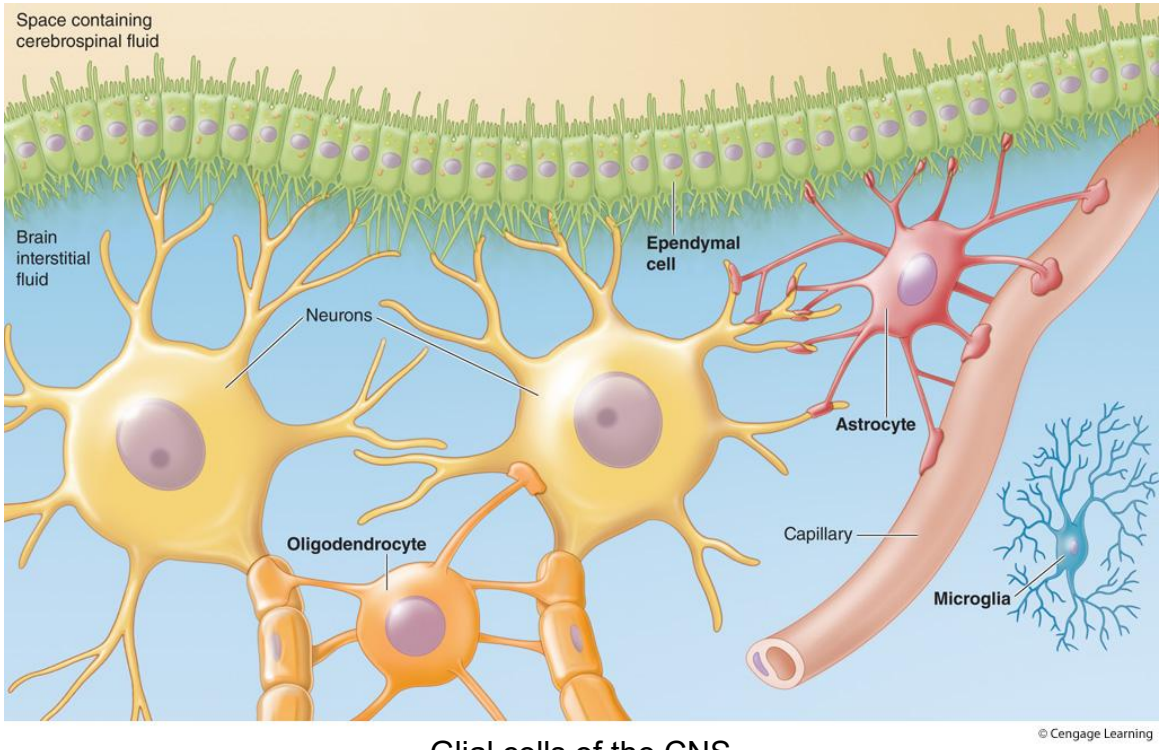
* Efferent autonomic nerve pathways consist of a two-neuron chain between the CNS and the effector organ.

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Glial cells of the CNS

- Glial cells do not initiate or conduct impulses
- Communicate with neurons and between themselves by chemical signals

- Glial cells support interneurons physically, metabolically, and functionally
  - Astrocytes
  - Oligodendrocytes
  - Microglia
  - Ependymal cells



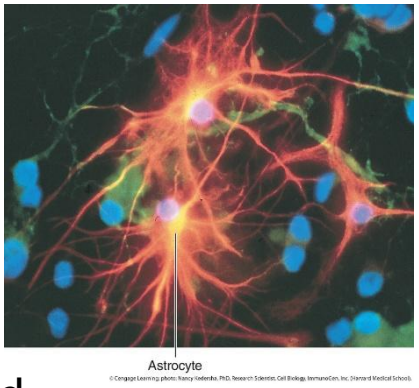
Glial cells of the CNS

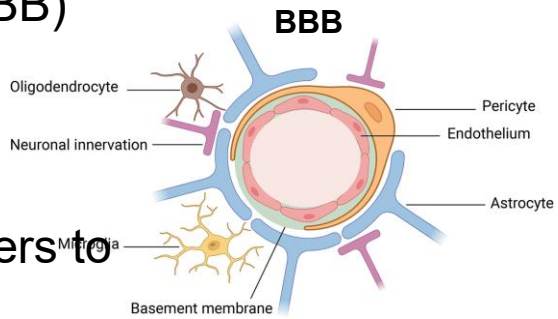
Primary functions of glial cells

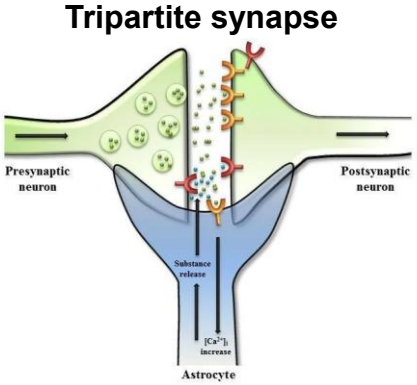
- Structural support
- Myelin production
- Waste cleanup
- Immune response

Glial cells of the CNS

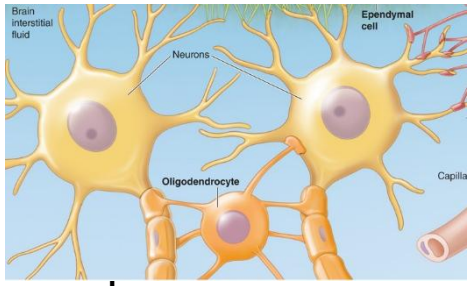
- Astrocytes: star-shaped, most abundant glial cells
  - Main glue, providing structural support, helping maintain proper neuronal organization
  - Act as scaffold guiding neurons to their proper final destination during fetal brain development
  - Induce capillaries of the brain to undergo the anatomic and functional changes to form the blood-brain barrier (BBB)
  - Help transfer nutrients from the blood to the neurons
  - Regulate extracellular K^+ concentration to maintain neuronal excitability
  - Take up and degrade locally released neurotransmitters to control the activity of chemical messengers
  - Participate in synaptic signaling (tripartite synapse)
  - Provide metabolic support (e.g., lactate) to neurons
  - Help transfer nutrients from blood to neurons
  - Form glial scars help repair brain injuries





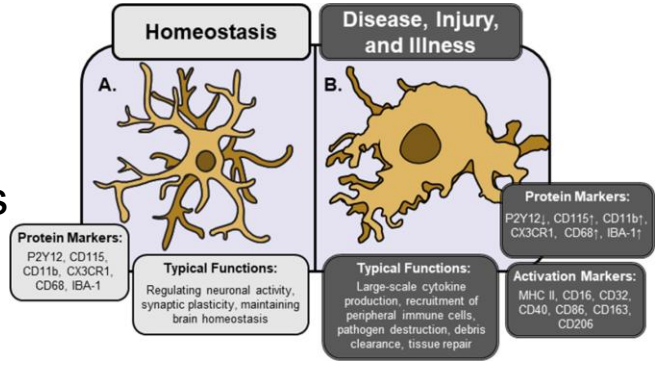


Glial cells of the CNS



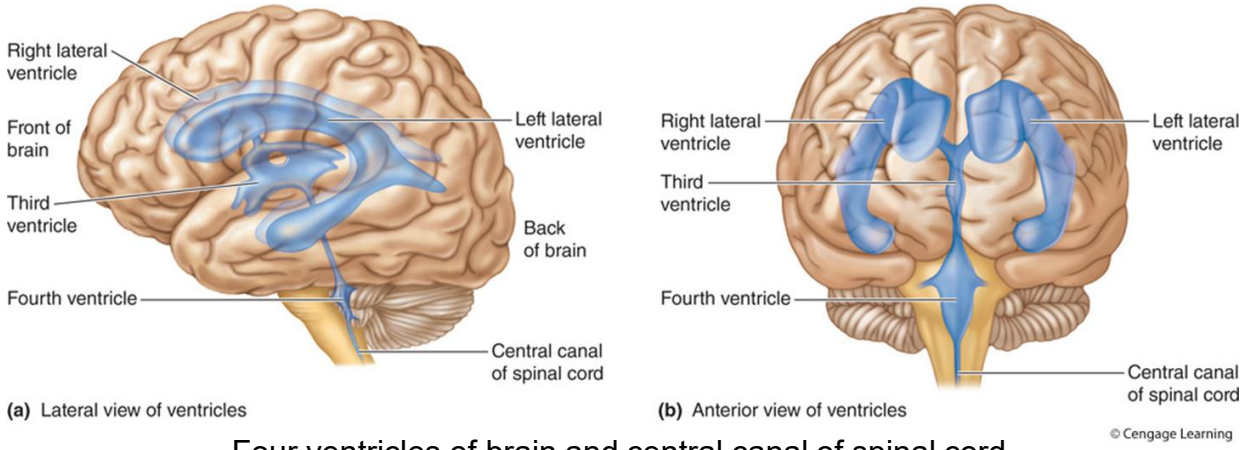
- Oligodendrocytes
  - Form insulative myelin sheaths around axons of the CNS
  - Multiple projections from oligodendrocytes wraps interneuronal axon

- Microglia
  - Immune defense cells similar to monocyte
  - Under resting, they release nerve growth factor (NGF), which help neurons and glial cells survive and proliferate
  - Participate in synapse pruning, process eliminating unneeded synapses, during development and memory formation
  - Under trouble (infection or injury), they round up and become highly mobile and move toward infected region to remove invaders or cell debris
  - Activated microglia also release destructive chemicals for offense against their target



Glial cells of the CNS

- Ependymal cells
  - Specialized ciliated neuroepithelial cells forming the lining of the brain's ventricular system and the spinal cord's central canal
  - Specialize in production, movement, and exchange of cerebrospinal fluid (CSF)
  - Act as the interface between brain tissue and CSF
  - Also serve as neural stem cell forming other glial cells and even neurons after nerve injury



Four ventricles of brain and central canal of spinal cord

- Unlike neurons, glia keep cell division -> prone to brain tumor

Glial cells (glia) in the nervous system



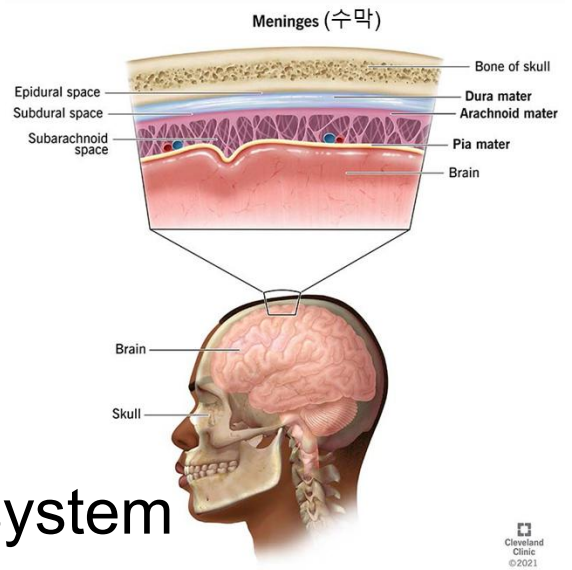
https://www.youtube.com/watch?v=AwES6R1_9PM

5.2 Protection and Nourishment of the Brain

Protection and nourishment of the brain

- Tissues of CNS is delicate, therefore damaged nerve cells cannot be replaced -> they need to be protected

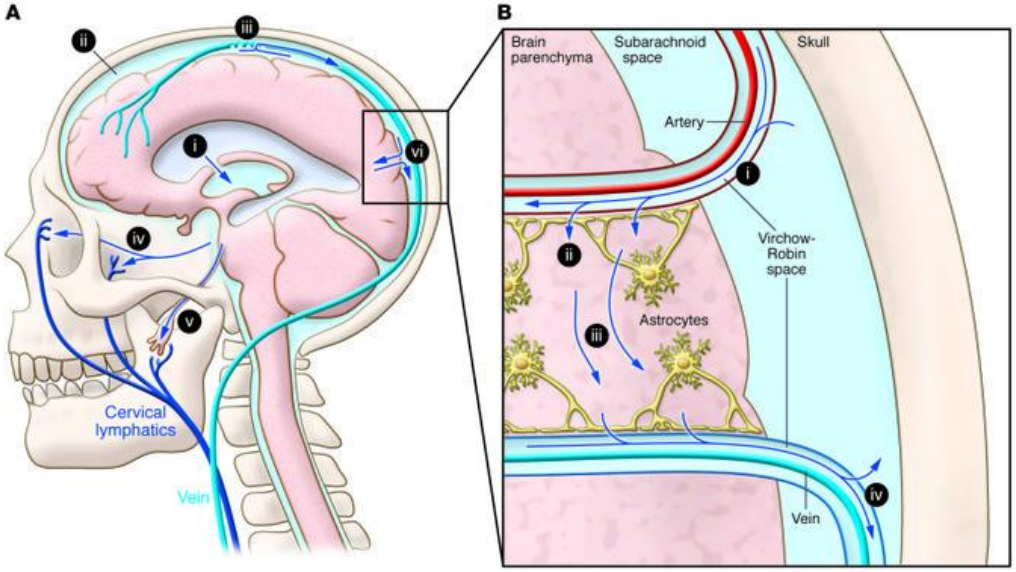
- Four major features protect the CNS from injury
  - Cranium and vertebral column
  - Meninges
  - Cerebrospinal fluid
  - Blood–brain barrier



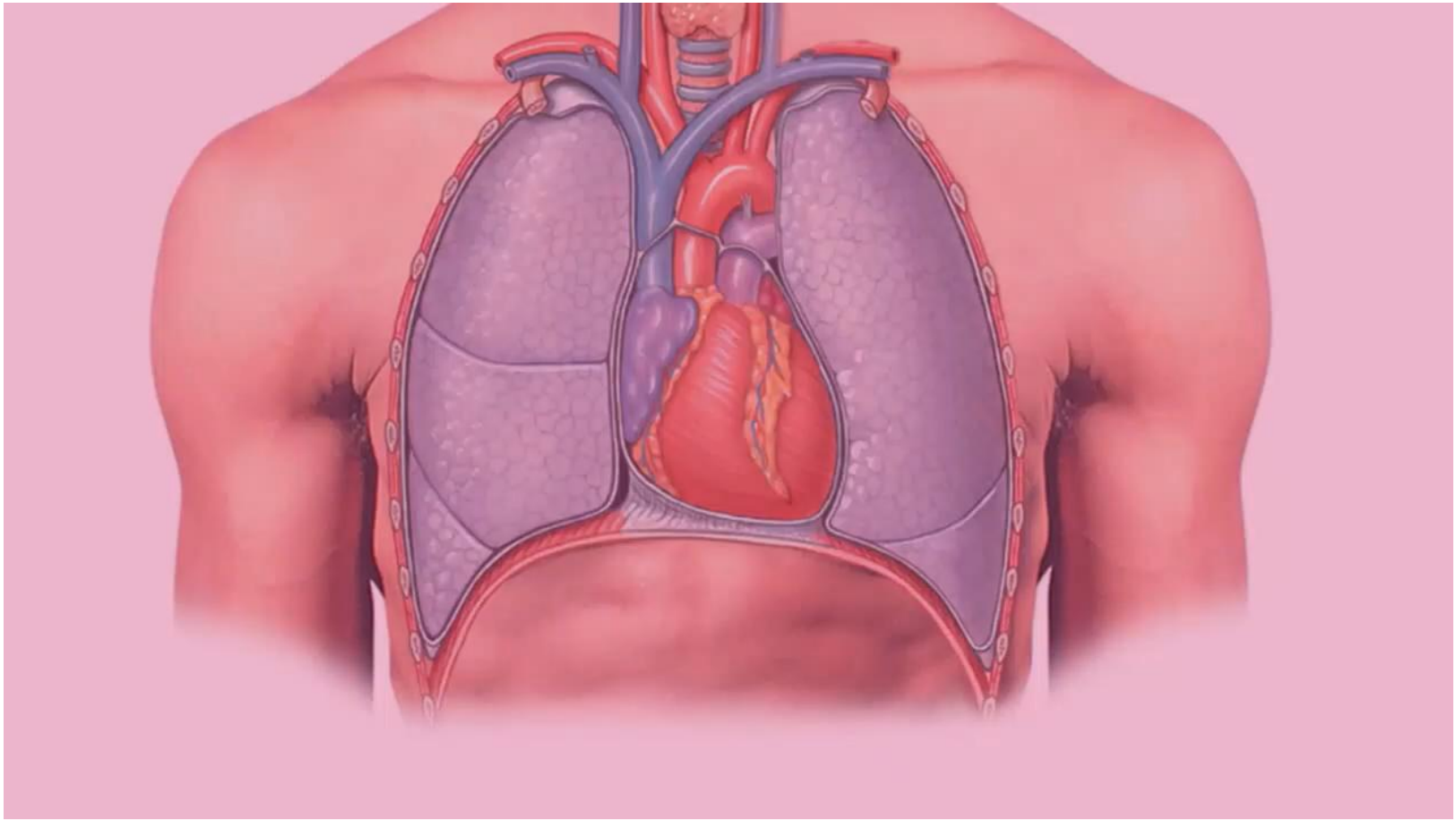
- Three meningeal membranes <u>wrap, protect, and nourish</u> the central nervous system
  - Dura mater, arachnoid mater, pia mater

The fluids of the brain

- The brain floats in its own special cerebrospinal fluid
  - CSF surrounds and cushions the brain and spinal cord
  - Exchange of materials between cells and interstitial fluid surrounding brain
  - CSF is formed by choroid plexuses of the ventricles
- The material exchange between the blood and brain via CSF is tightly controlled by highly selective blood-brain barrier
  - BBB protects the CNS from harmful changes and limits access of blood-borne materials into brain



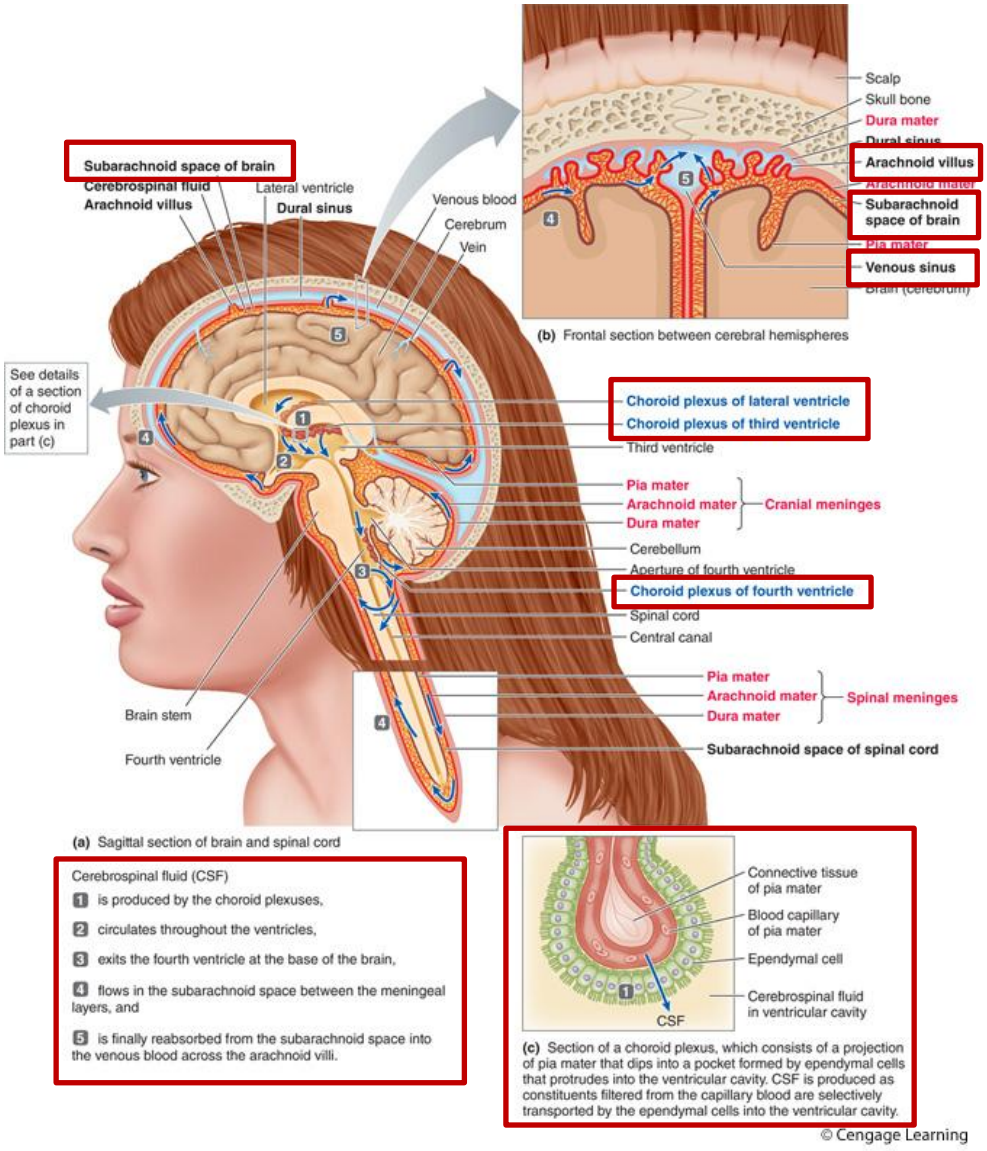
Formation and circulation of CSF



Main players: four ventricles, choroid plexus, interstitial fluid, subarachnoid space, lateral/median aperture, arachnoid granulation

https://www.youtube.com/watch?v=FvnoS1TCUCk

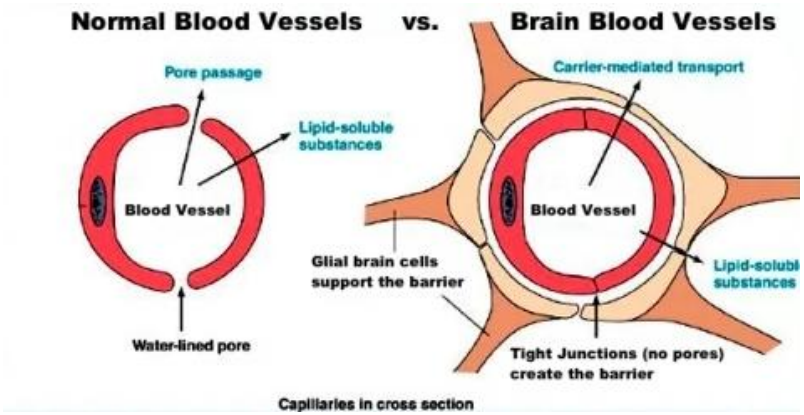
The fluids of the brain

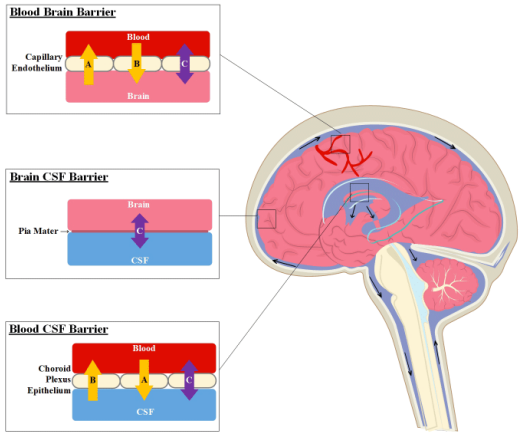


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Blood-brain barrier (BBB)

- BBB protects the brain from chemical fluctuations in blood
  - Strictly limits exchange between blood and brain
  - Minimizes possibility that harmful blood-borne substances reaching CNS
  - Prevents certain circulating hormones that could also act as neurotransmitters from reaching the brain
  - Limits use of drugs for treatment of the brain and spinal cord disorders -> many drugs cannot penetrate BBB





The nourishment of the brain

- The brain depends on constant delivery of oxygen and glucose by the blood
  - Unlike most tissues capable of anaerobic respiration, brain cannot produce ATP without O_2
  - Unlike other tissues, brain cannot store glucose
  - Brain uses 20% of oxygen and 50% of glucose consumed in the body under resting condition

  - Brain damage results if deprived of O_2 or glucose by blood (stroke)
  - Oxygenated blood is transported through tiny capillaries, where oxygen is released and glucose is moved across the blood-brain barrier via GLUT1 transporters

5.3 Overview of the Central Nervous System

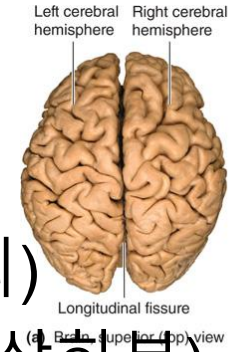
Overview of the CNS

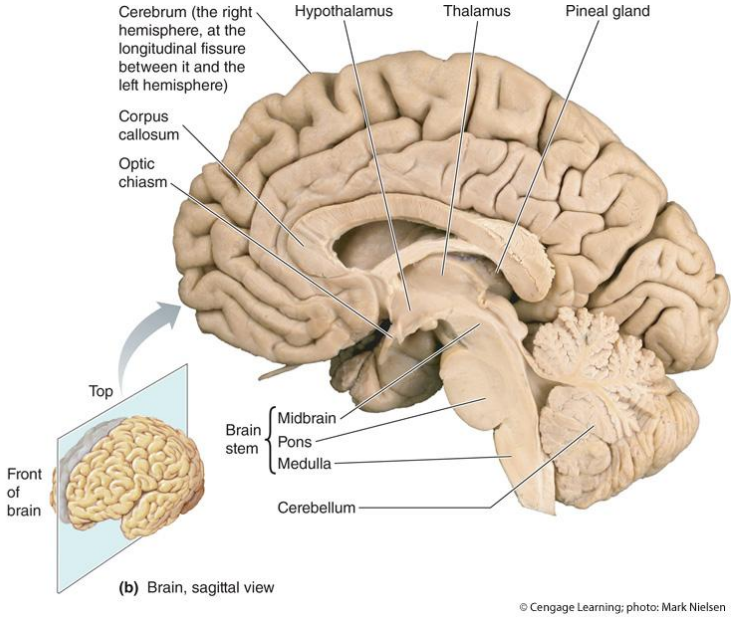
- The CNS consists of the brain and spinal cord: 85 billion neurons in brain joined by synapses

- Enable you to (1) regulate internal environment, (2) experience emotions, (3) voluntarily control movements, (4) perceive body and surroundings, and (5) engage in higher cognitive processes (thinking, memory)

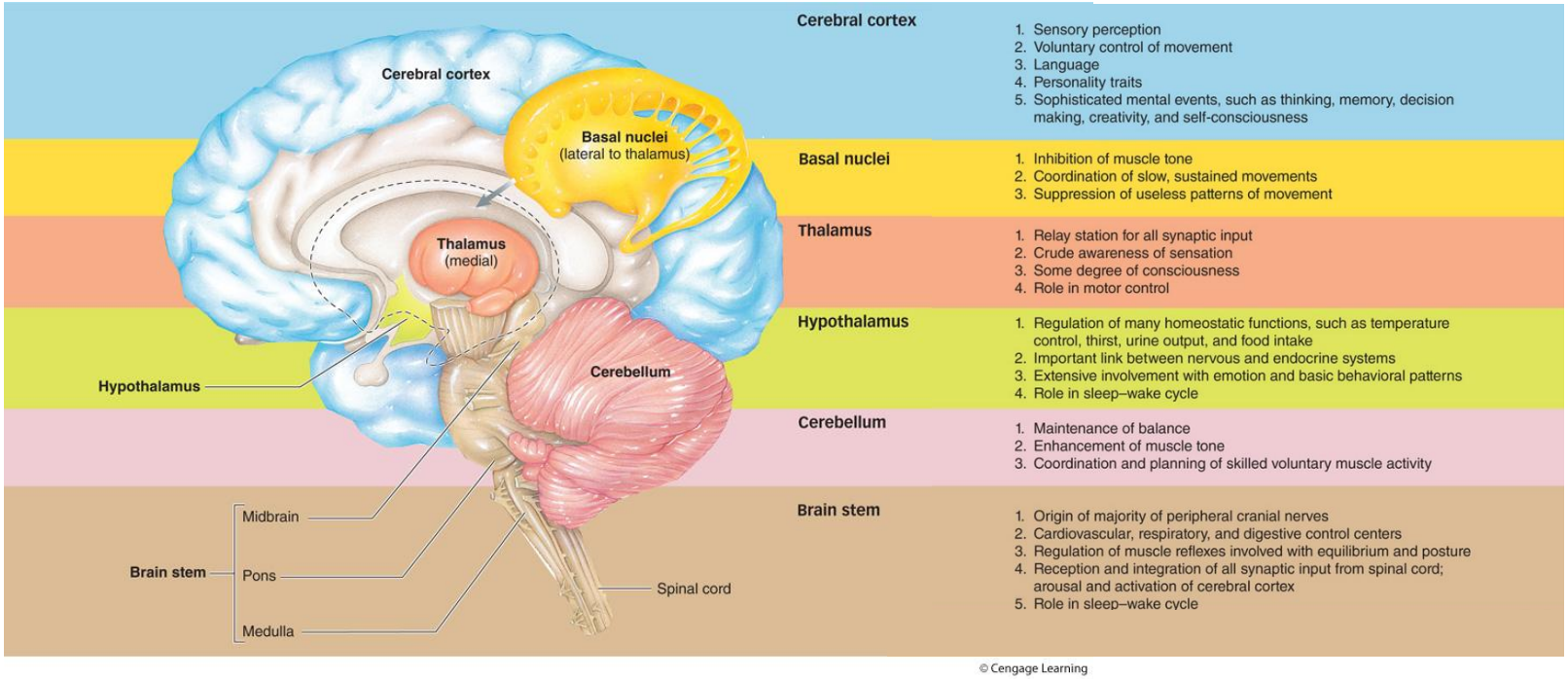
Brain components

- Brain stem (뇌간)
- Cerebellum (소뇌)
- Forebrain (앞뇌)
  - Diencephalon (사이뇌)
    - Hypothalamus (시상하부)
    - Thalamus (시상)
  - Cerebrum (대뇌)
    - Basal nuclei (기저핵)
    - Cerebral cortex (대뇌피질)





Brain components



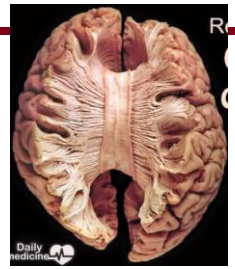
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Brief summary

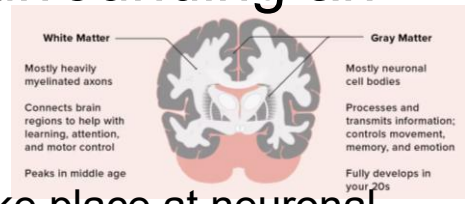
- Brain stem: midbrain, pons, medulla (life-sustaining processes)
- Cerebellum: proper body position, coordination of movement
- Forebrain
  - Diencephalon
    - Hypothalamus: homeostatic functions maintaining stability of internal environment
    - Thalamus: begins sensory processing
  - Cerebrum (found in advanced vertebrates)
    - Basal nuclei (inner core): regulate voluntary muscle movement
    - Cerebral cortex (outer layer of cerebrum): voluntary initiation of movement, final sensory perception, thought, language, mind

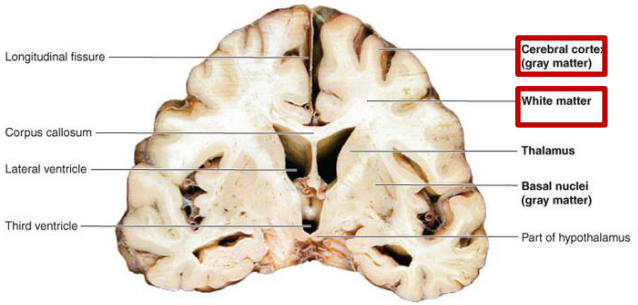
5.4 Cerebral Cortex (대뇌피질)

Cerebrum and cerebral cortex



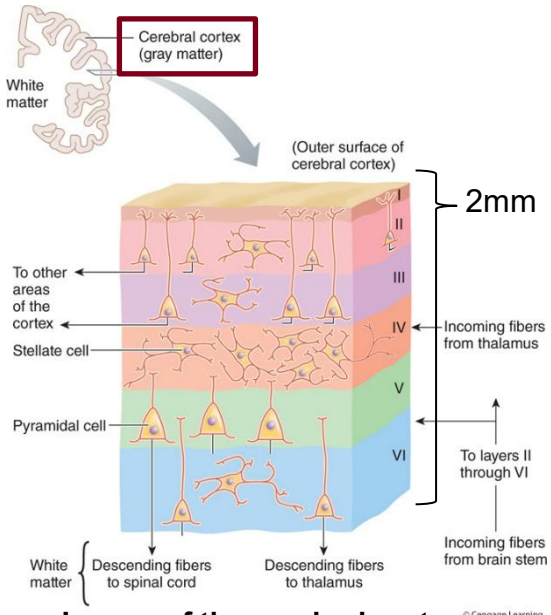
- Cerebrum is divided into left and right cerebral hemispheres, connected by corpus callosum
  - Corpus callosum: thick band consisting of approx. 300 million neuronal axons connecting two hemispheres
- Cerebral cortex: outer shell of gray matter, surrounding an inner core of white manner
  - Gray matter: processing center of the CNS
    - Integration of neural input and initiation of neural output take place at neuronal cell bodies and synapses in gray matter
  - White matter: wires connecting computers, communication pathway
    - Fibers in white matter transmit signals within the cerebral cortex or to other regions of CNS to facilitate integration of activity
    - Mostly myelinated axons
    - Not cerebral cortex





Cerebral cortex and neuronal network

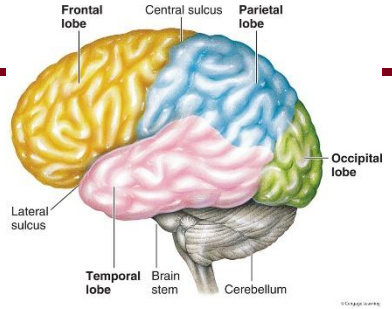
- Cerebral cortex is organized into layers and functional columns
  - It is organized into six layers based on varying distribution of distinctive cell types
  - Neurons in same column function as a team
  - Layer I is mostly glial cells and axons that run laterally.
  - Layers II through VI contain different proportions of two main classes of cortical neurons: pyramidal cells, which are the major output neurons, and stellate cells, which primarily receive input to the cortex and process local information



Layers of the cerebral cortex

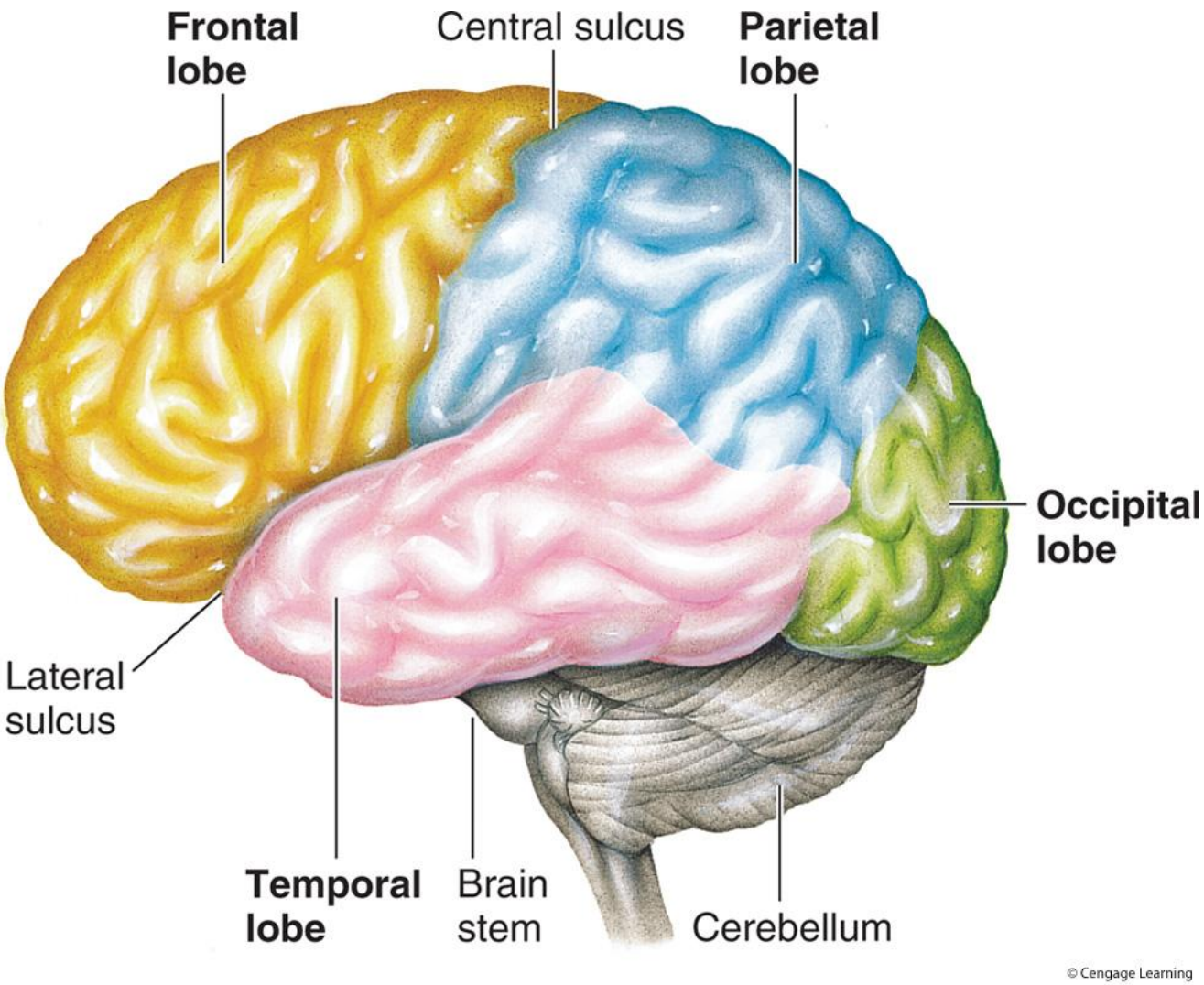
  - Functional differences between cortex areas result from different layering patterns and different input–output connections, not from unique cell type or neuronal mechanism

Lobes of the cerebral cortex



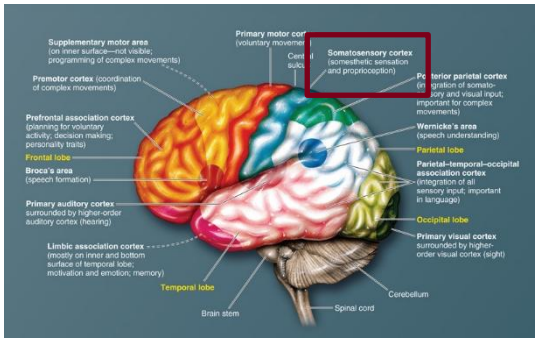
- The four pairs of lobes in the cerebral cortex are specialized for different activities
  - Keep in mind that even though discrete activity is attributed to a particular region of the brain, each part depends on complex interplay among other various regions for moving in/out messages
  - Anatomically each half of cortex is divided into..
  - Occipital lobes (후두엽) house the visual cortex
  - Temporal lobes (측두엽) house the auditory cortex
  - Parietal lobes (두정엽) are responsible for receiving and perceiving somatosensory inputs (general sensory inputs except for special senses; pain, temperature, proprioception)
  - Frontal lobes (전두엽) are responsible for voluntary motor movement, speaking, elaboration of thought
  - Central sulcus: deep infolding separating parietal and frontal lobes

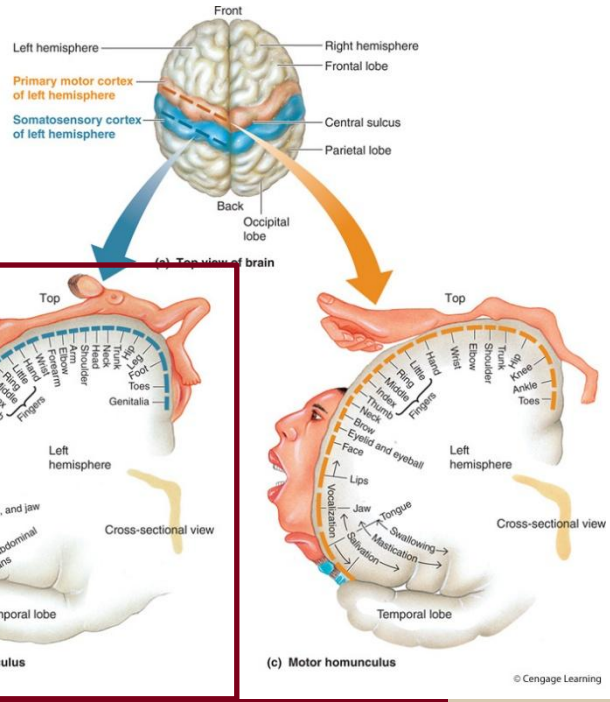
Lobes of the cerebral cortex



Parietal lobes

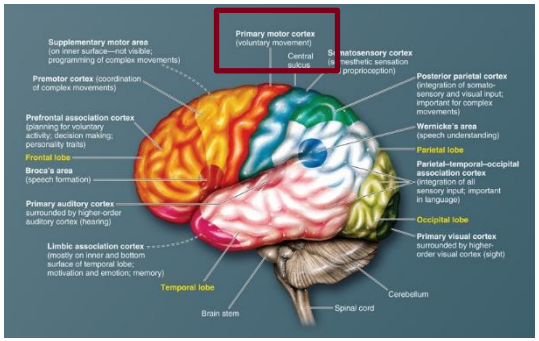
- Parietal lobes accomplish somatosensory processing
  - Sensations from the surface of the body, including touch, pressure, heat, cold, and pain
  - These information is detected by sensory receptors in the skin and relayed along afferent fibers to the CNS
  - Within the CNS, the information is projected to somatosensory complex in parietal lobes
  - Somatosensory cortex receives sensory inputs from the opposite side

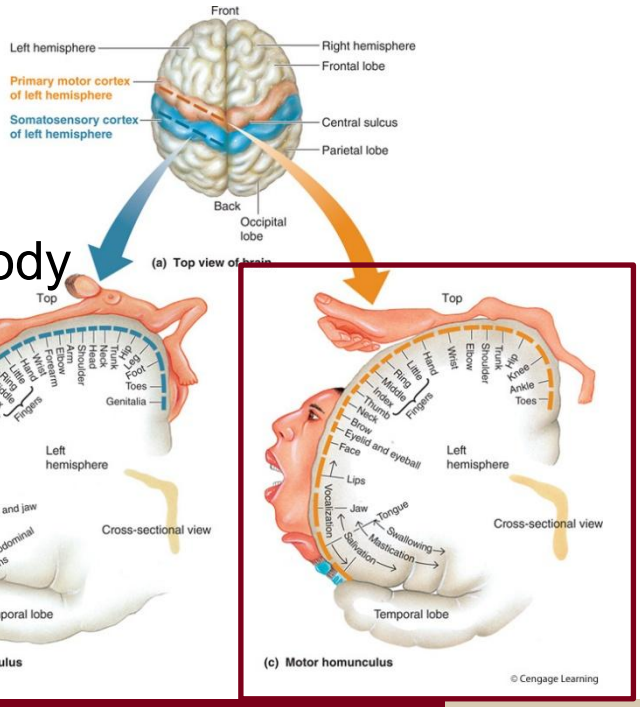




Frontal lobes

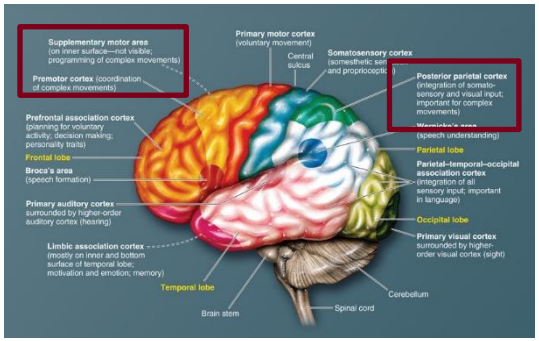
- The primary motor cortex located in the frontal lobes controls voluntary control over movement produced by skeletal muscles
  - Neuronal tracts originating in the motor cortex of left hemisphere cross over before passing down the spinal cord to reach efferent motor neurons triggering muscle contraction on the right side sensory inputs from the opposite side
  - Stimulation of different areas of the primary motor cortex brings about movement in different regions of the body

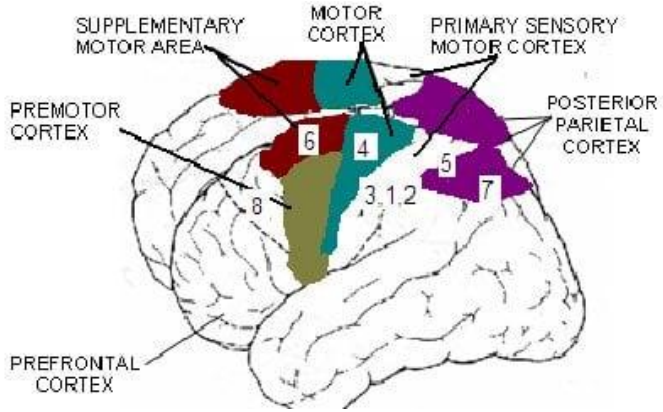




Frontal and parietal lobes

- Primary motor cortex is not the only region involved in motor control
- Higher motor areas are also important in motor control
  - Supplementary motor area: in front of primary motor cortex, invisible, preparatory role in programming complex movement
  - Premotor cortex: in front of primary motor cortex, orienting the body and arms toward a specific target
  - Posterior parietal cortex: posterior to somatosensory cortex, guides premotor cortex

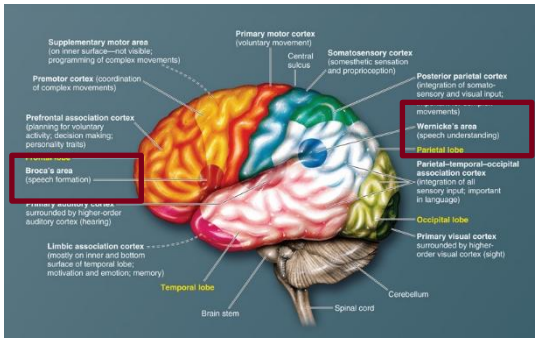




Main parts of the CNS involved in motor control

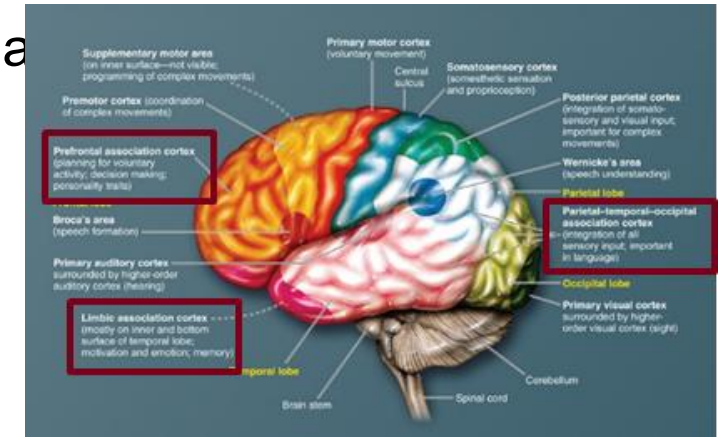
Plasticity and language in the cerebral cortex

- Plasticity: ability to change or be functionally remodeled in response to demands
  - The brain can be remodeled in response to various demands
- Different regions of the cortex control different aspects of language
  - Broca's area: located in left frontal lobe near motor areas, controls muscles necessary for speaking
  - Wernicke's area: located in the left cortex at the intersection of parietal, temporal, and occipital lobes, associated with language comprehension of both spoken and written messages

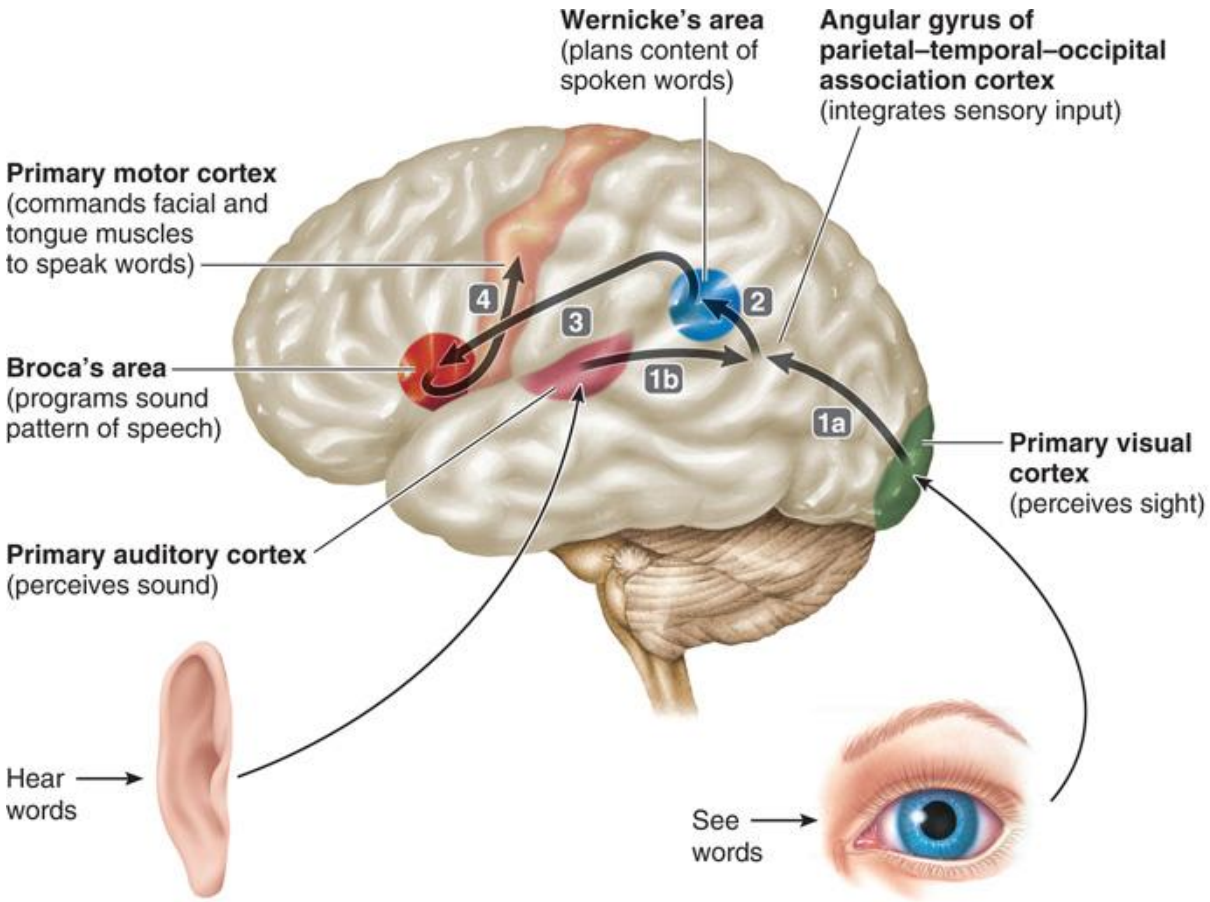


The association area in the cerebral cortex

- The association area: do not directly process sensory inputs, but is involved in many higher functions
  - Prefrontal association cortex: brainstorming and thinking. Especially, planning for voluntary activity, decision making, creativity, and personality traits
  - Parietal-temporal-occipital association cortex: pools and integrates somatic, auditory, and visual senses for complex perceptual processing, involved in language pathway
  - Limbic association cortex: joining inner portion of each temporal lobe, motivation/emotion/memory
  - Total cerebral cortex = motor/sensory/la association areas (연합영역)



Cortical pathway for speaking a word seen or heard



1a To speak about something seen, the brain transfers the visual information from the primary visual cortex to the angular gyrus of the parietal–temporal–occipital association cortex, which integrates inputs such as sight, sound, and touch.

1b To speak about something heard, the brain transfers the auditory information from the primary auditory cortex to the angular gyrus.

2 The information is transferred to Wernicke's area, where the choice and sequence of words to be spoken are formulated.

3 This language command is then transmitted to Broca's area, which translates the message into a programmed sound pattern.

4 This sound program is conveyed to the precise areas of the primary motor cortex that activate the appropriate facial and tongue muscles for causing the desired words to be spoken.

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Hearing Questions ➡ [ ] ➡ [ ] ➡ [ ] ➡ [ ] ➡ Primary motor cortex

Specialization in the cerebral cortex

- Cerebral hemispheres have some degree of specialization
  - Left cerebral hemisphere: logical, analytical, sequential, and verbal tasks
  - Right cerebral hemisphere: non-language skills

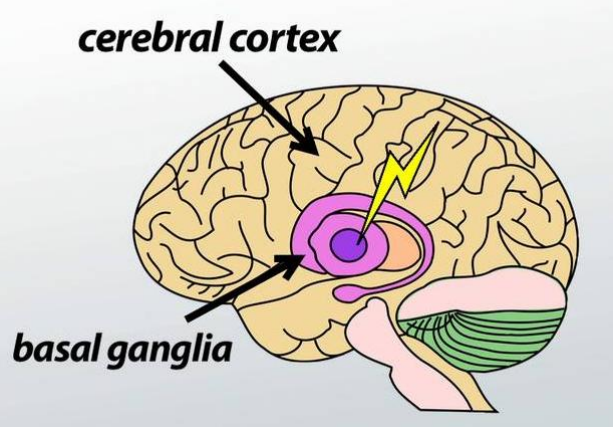
- Cerebral lateralization: the tendency for cognitive functions to be specialized to either the left or right hemisphere of the brain

5.5 Basal Nuclei (기저핵), Thalamus (시상), And Hypothalamus (시상하부)

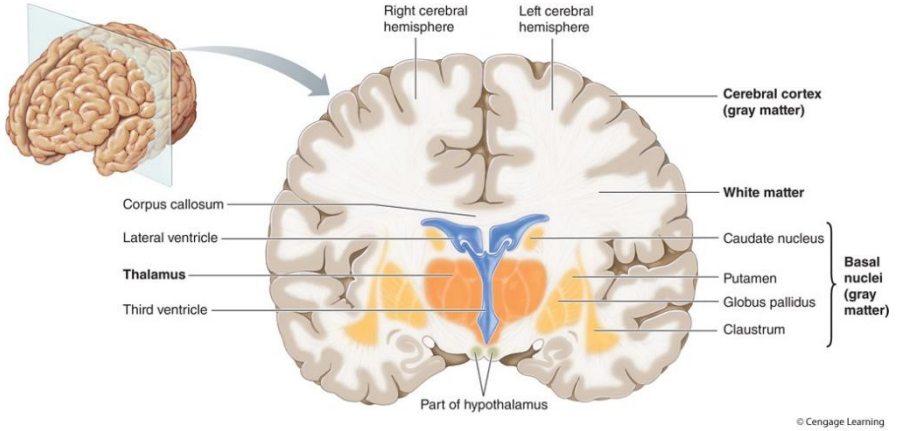
Basal nuclei

- Basal nuclei (basal ganglia) is a cerebrum component, consisting of several masses of gray matter
  - Located deep within the cerebral white matter
  - It plays an important inhibitory role in motor control
  - (1) Inhibits muscle tone (근긴장), (2) selectively maintain purposeful motor activity, (3) monitors and coordinates slow and sustained contractions (posture and support)

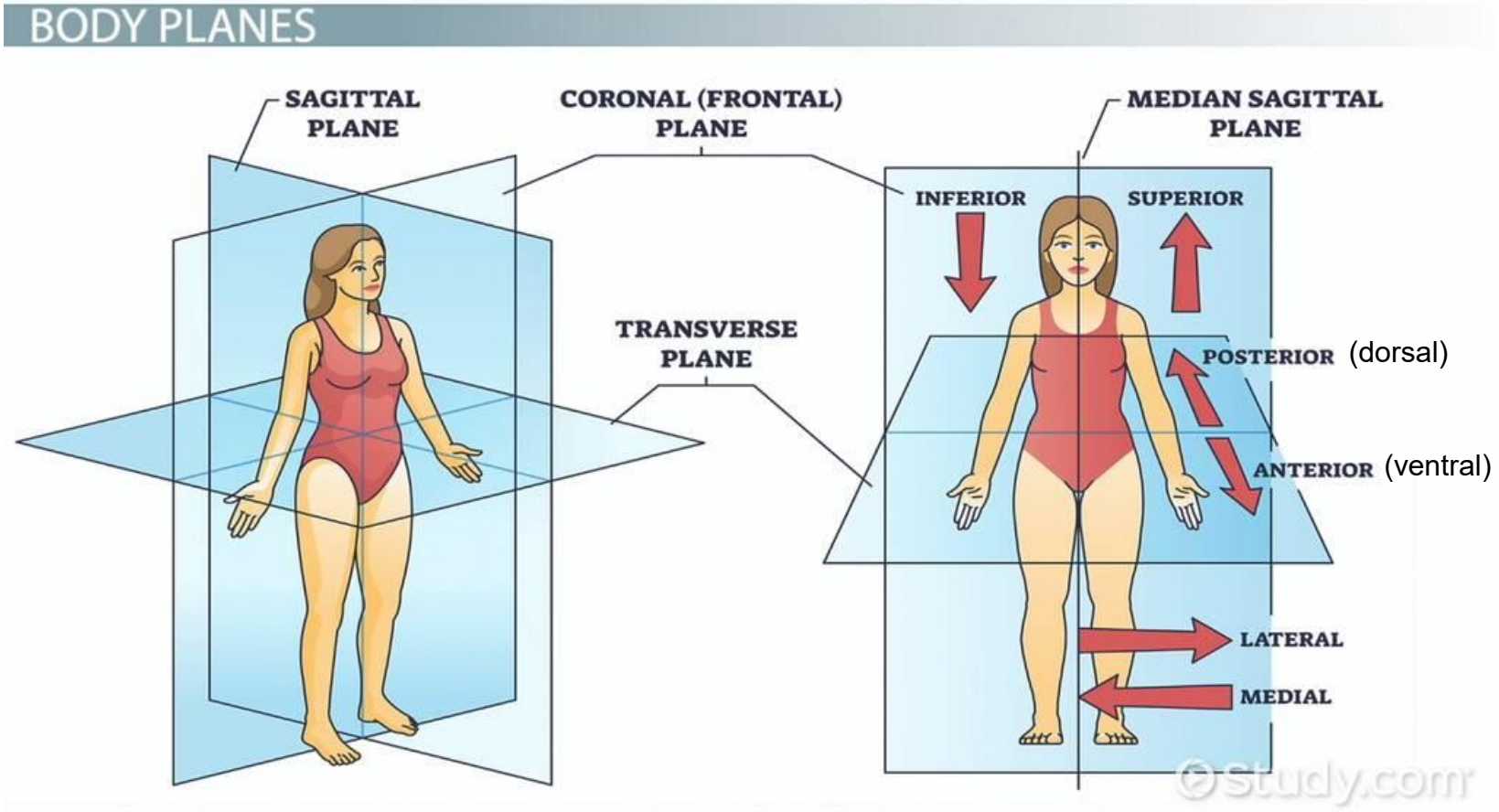
Sagittal section of the brain



Frontal section of the brain

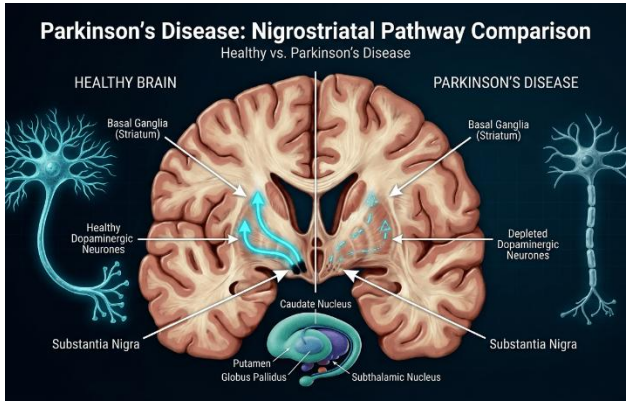


Body planes


BODY PLANES
SAGITTAL PLANE
CORONAL (FRONTAL) PLANE
MEDIAN SAGITTAL PLANE
TRANSVERSE PLANE
INFERIOR
SUPERIOR
POSTERIOR (dorsal)
ANTERIOR (ventral)
LATERAL
MEDIAL
©Study.com

Basal nuclei and Parkinson's disease

- Parkinson's disease
  - The importance of basal nuclei in motor control
  - Destruction of neurons releasing neurotransmitter dopamine in the basal nuclei
  - Since basal nuclei lacks enough dopamine to exert their normal function..
    - Increased muscle tone or rigidity
    - Involuntary, useless, unwanted muscle movement
    - Slowness in carrying out motor behaviors



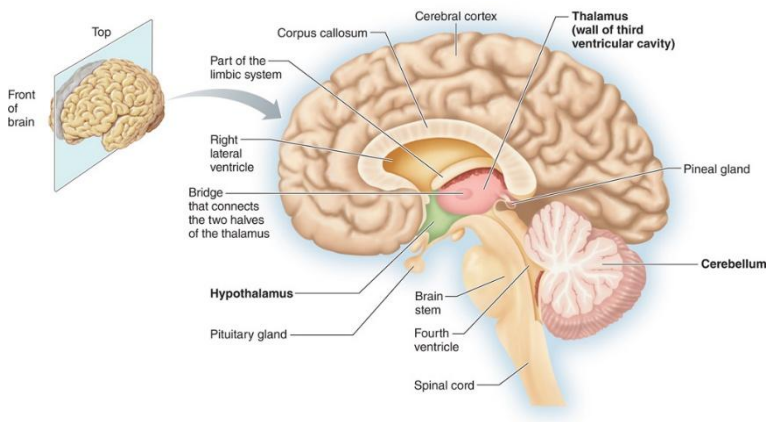


  - As a treatment, dopamine precursor (levodopa) is given
  - Dopamine itself cannot penetrate the BBB

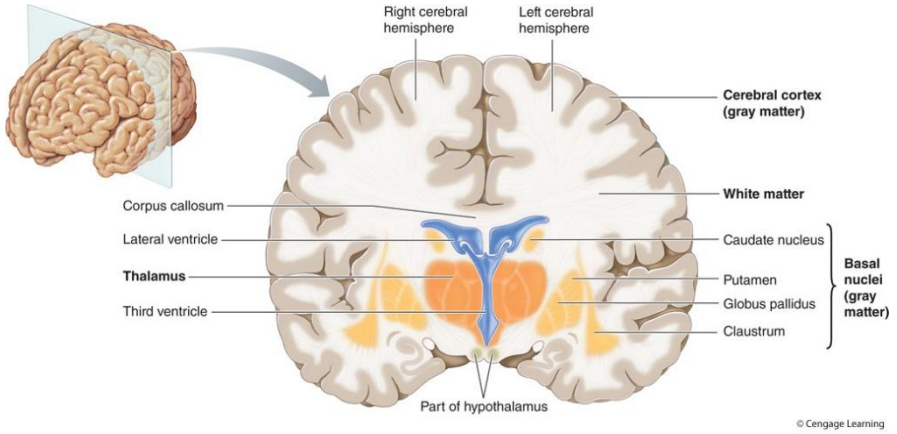
The thalamus and hypothalamus (diencephalon)

- The thalamus is a sensory relay station and is important in motor control
  - "Relay station" for preliminary processing of sensory input
  - On the way of sensory input synapses to the cortex, thalamus screens out unnecessary signals and guides the important signals to appropriate cortex area
  - Helps direct attention to stimuli of interest (traffic outside vs baby)

Sagittal section of the brain



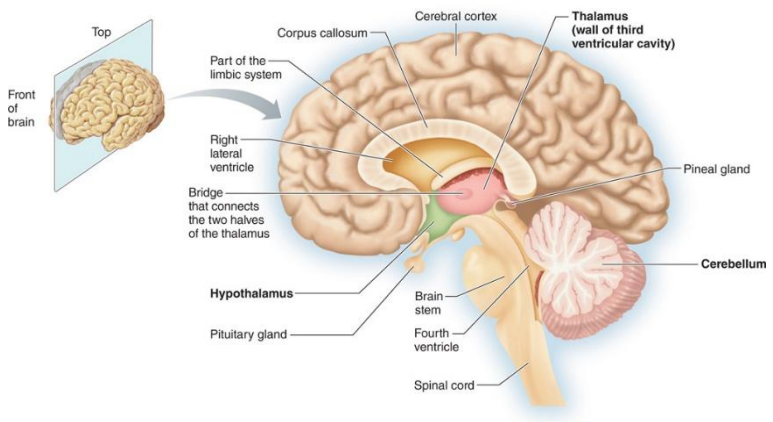
Frontal section of the brain



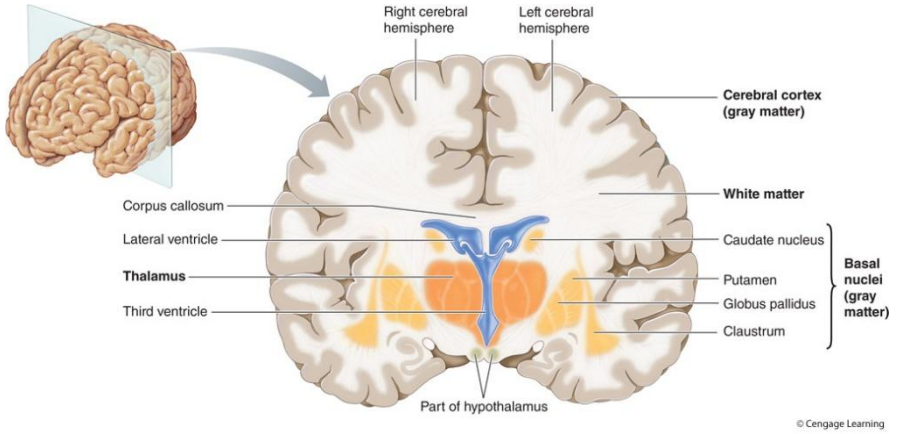
The thalamus and hypothalamus (diencephalon)

- The hypothalamus regulates many homeostatic functions
  - Collection of specific nuclei and associated fibers that lie beneath the thalamus
  - Integrating center for many important homeostatic functions
  - Links between autonomous nervous system (ANS) and endocrine
    - Body temperature, thirst and urine output, food intake, anterior pituitary (뇌하수체전엽) hormone secretion, posterior pituitary hormone production, uterine contraction and milk ejection, ANS controlling center, emotional and behavior pattern, sleep-wake cycle (INTERNAL ENVIRONMENT REGULATION)

Sagittal section of the brain



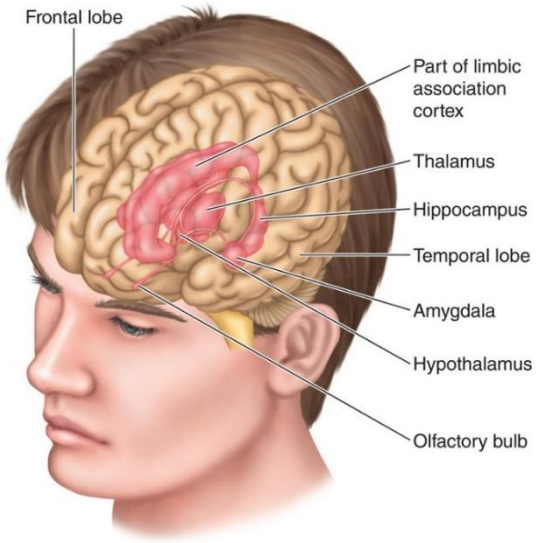
Frontal section of the brain



5.6 Emotion, Behavior, and Motivation - Limbic system (변연계)

The limbic system

- The limbic system is a functional system
  - Not a separate structure, but a ring of forebrain structures surrounding the brain stem
  - Includes the portions of the cerebral cortex (cingulate gyrus and parts of the prefrontal cortex), the amygdala, hippocampus, hypothalamus, thalamus, olfactory bulbs, basal nuclei, etc)
  - Responsible for emotions, basic behavioral patterns related to survival and perpetuation of the species, motivation, learning, and memory



The limbic system



https://www.youtube.com/watch?v=LNs9ruzoTmI

The limbic system

Major components	Function in limbic system
Cingulate gyrus	emotional processing and regulation
Prefrontal cortex	decision-making, emotional regulation
Thalamus	relays limbic information and participates in emotional and memory circuits
Hypothalamus	bodily homeostasis (hunger, thirst, sleep, body temperature) and hormone production
Hippocampus	long-term memory formation and spatial navigation
Amygdala	processing emotions, particularly fear, anxiety, and anger, and managing fight-or-flight responses
Septal nuclei	reward and reinforcement
Basal nuclei	reward and motivated behavior
Olfactory bulbs	processes smells and links them to memories/emotions

Emotion, behavior, and neurotransmitters

- The underlying neurophysiological mechanisms responsible for the psychological observations of emotions and motivated behavior largely remain a mystery
- Norepinephrine, dopamine, and serotonin are neurotransmitters in pathways for emotions and behavior, and they are involved in some way
- All effective anti-depressant drugs increase the availability of *norepinephrine and serotonin* in the CNS
  - Selective serotonin reuptake inhibitors (SSRIs)
  - Serotonin norepinephrine reuptake inhibitors (SNRIs)
  - Tricyclic antidepressants (TCAs)
  - Monoamine oxidase inhibitors (MAOIs)

5.7 Learning and Memory
- Cerebellum (소뇌)

Learning and memory

- Learning: acquisition of knowledge from experience
  - Rewards and punishments are integral parts
  - More or less likely to repeat the same response next time

- Memory: storage of acquired knowledge, laid down in stages
  - From learning and memory, the individual adapt their behavior to particular external circumstances
  - Short-term memory lasts for second to hours
  - Long-term memory is retained for days to years
  - Working memory (작용 기억): short-term memory temporarily holding and manipulating information for complex tasks like learning, reasoning, and comprehension

consolidation

Short-term and long-term memory involve different molecular mechanisms

TABLE 5-2 Comparison of Short-Term and Long-Term Memory

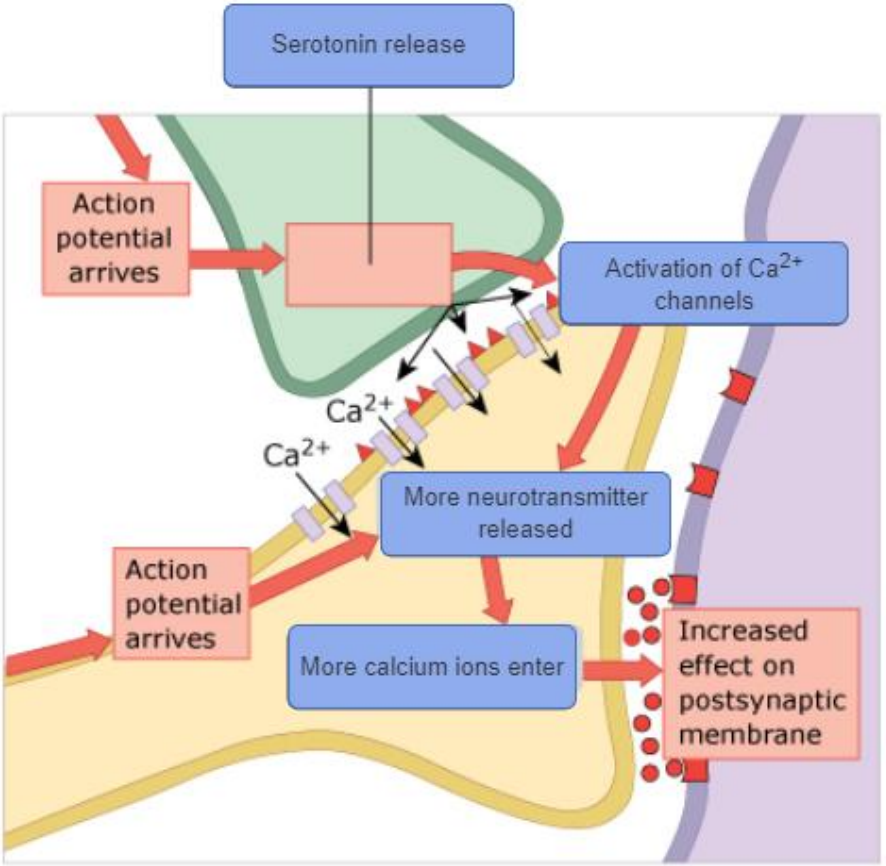
Characteristic	Short-Term Memory	Long-Term Memory
Time of storage after acquisition of new information	Immediate	Later; must be transferred from short-term to long-term memory through consolidation; enhanced by practice or recycling of information through short-term mode
Duration	Lasts for seconds to hours	Retained for days to years
Capacity of storage	Limited	Very large
Retrieval time (remembering)	Rapid retrieval	Slower retrieval, except for thoroughly ingrained memories, which are rapidly retrieved
Inability to retrieve (forgetting)	Permanently forgotten; memory fades quickly unless consolidated into long-term memory	Usually only transiently unable to access; relatively stable memory trace
Mechanism of storage	Involves transient modifications in functions of preexisting synapses, such as altering amount of neurotransmitter released	Involves relatively permanent functional or structural changes between existing neurons, such as formation of new synapses; synthesis of new proteins plays a key role

Short-term memory involves transient changes in preexisting synaptic activity

- Habituation (습관화) and sensitization (민감화)
  - Results from modification of different channel proteins in presynaptic terminals mediating the behavior being modified
  - Habituation: decreased responsiveness to repetitive presentation of indifferent stimulus (no reward or punish)
    - Ignoring sound of ticking clock or being accustomed to smell of new room
    - Purpose: Helps conserve energy by ignoring irrelevant stimuli
    - Reduce synaptic activity by presynaptic voltage-gated Ca^{2+} channels decreasing the opening as an AP arrival, leading to decreased NT release and diminished synaptic strength
  - Sensitization: increased responsiveness to mild stimuli after strong or noxious stimulus
    - A person with trauma have increased startle response to loud noises, or becoming more irritated by a dripping faucet over time
    - Purpose: Prepares the organism to respond to threats
    - Enhance synaptic activity by indirectly enhancing Ca^{2+} entry via serotonin-mediated presynaptic facilitation, leading to increased NT release and strengthened synaptic strength

Short-term memory involves transient changes in preexisting synaptic activity

- Presynaptic facilitation
  - A short-term form of synaptic plasticity where a second nerve impulse produces a larger postsynaptic response when it closely follows a first one
  - Modulatory neurotransmitter prolongs action potential and increase neurotransmitter release from presynaptic neuron
  - Underlying mechanism of sensitization and its repeated activation contributes to memory by strengthening specific synapse

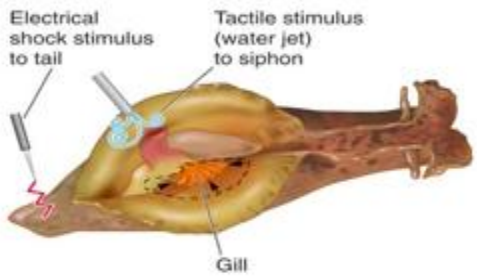


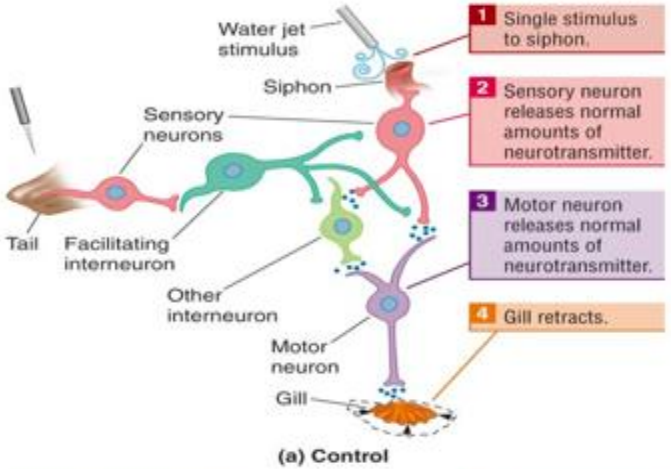
Presynaptic facilitation

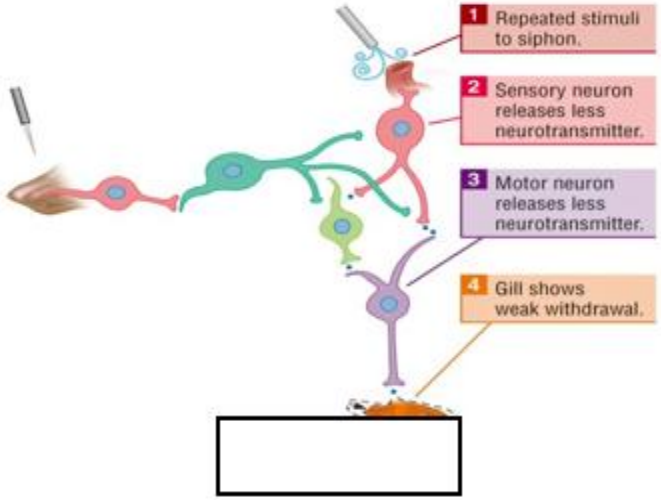
Kandel's research on marine snail *Aplysia californica*

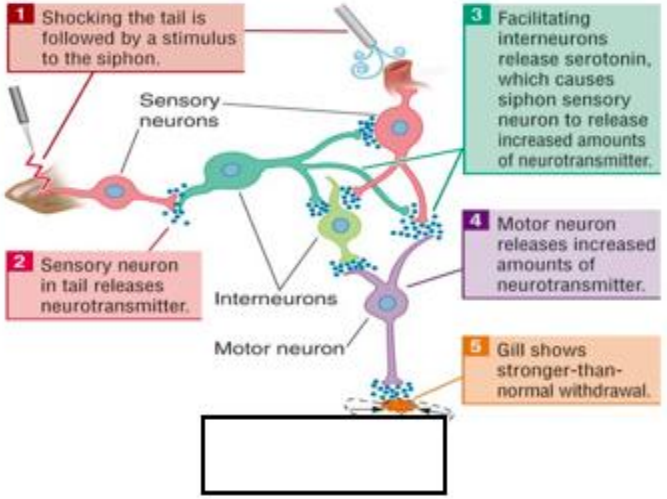


THE NOBEL PRIZE





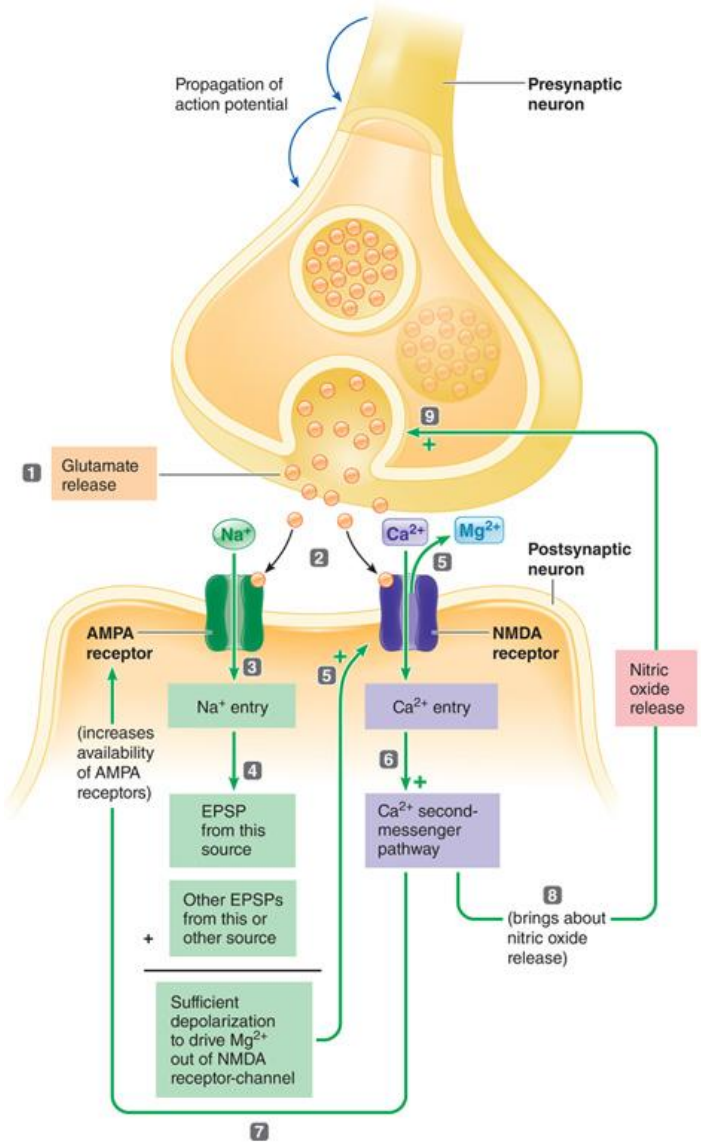




Short-term memory involves transient changes in preexisting synaptic activity

- Long-term potentiation
  - More complex memories are initially stored by long-term potentiation (LTP), a mechanism involving more persistent strengthening in activity of preexisting synapses
  - Connection gets stronger the more often it is used
  - Strengthening of synaptic activity: more EPSP formation in the postsynaptic neuron in response to presynaptic input
  - More action potential sent along this postsynaptic neuron than other neurons
  - Prevalent in hippocampus, converting short-term memories to long-term
  - Can occur by changes in both presynaptic and postsynaptic cells
    - Presynaptic: increased release of NT
    - Postsynaptic: increased responsiveness to NT from inserting more receptors
  - Synaptic strength: efficacy & magnitude of signal transmission between synapse

Long-term potentiation



AMPA receptor: chemically gated receptor-channel complex that opens on binding of glutamate. Permits Na^+ entry for the formation of EPSP.
NMDA receptor: chemically and voltage gated receptor-channel complex that opens upon both glutamate binding and postsynaptic depolarization. Permits Ca^{2+} entry for second messenger pathway and retrograde messenger (NO) production.

1. Glutamate is released from activated presynaptic neuron.
2. Glutamate binds with both AMPA and NMDA receptors.
3. Binding opens AMPA receptor-channel.
4. Na^+ entry through open AMPA channel depolarizes postsynaptic neuron, producing EPSP.
5. Binding opens gate of NMDA receptor-channel but Mg^{2+} still blocks channel. Sufficient depolarization from this AMPA opening plus other EPSPs drives Mg^{2+} out.
6. Ca^{2+} entry through open NMDA channel activates Ca^{2+} second-messenger pathway.
7. Second-messenger pathway promotes insertion of additional AMPA receptors in postsynaptic membrane, increasing its sensitivity to glutamate.
8. Second-messenger pathway also triggers release of retrograde paracrine (likely nitric oxide).
9. Nitric oxide stimulates long-lasting increase in glutamate release by presynaptic neuron.

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Long-term potentiation



Definition and significance of LTP, members and ions

https://www.youtube.com/watch?v=-mHgPfXHzJE

Long-term memory involves formation of new, permanent synaptic connections

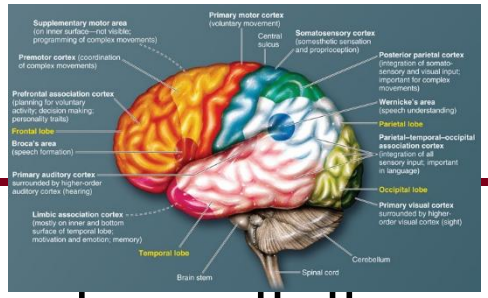
- Whereas short-term memory involves “transient” strengthening of “preexisting” synapse, long-term memory storage requires synthesis of proteins needed for long-lasting structural and functional synapse change
- Formation of new synapse or permanent changes in pre- or post-synaptic membrane to permanent physical change
- Players in long-term memory
  - cAMP responsive element binding protein (CREB): binds to DNA to influence synthesis of new proteins maintaining long-term memory
  - Immediate early genes (IRG): coding genes needed for structural changes or used for synthesis of more NTs or receptors
- Consolidation of STM to LTM
  - Information is transferred from STM to LTM via consolidation, a time-dependent process involving cellular changes in the brain

Memory traces (기억 흔적)

- Memory traces are present in multiple regions of the brain, not stored in a single brain location
  - Hippocampus and declarative memories
  - Cerebellum and procedural memories
  - Prefrontal cortex and working memory

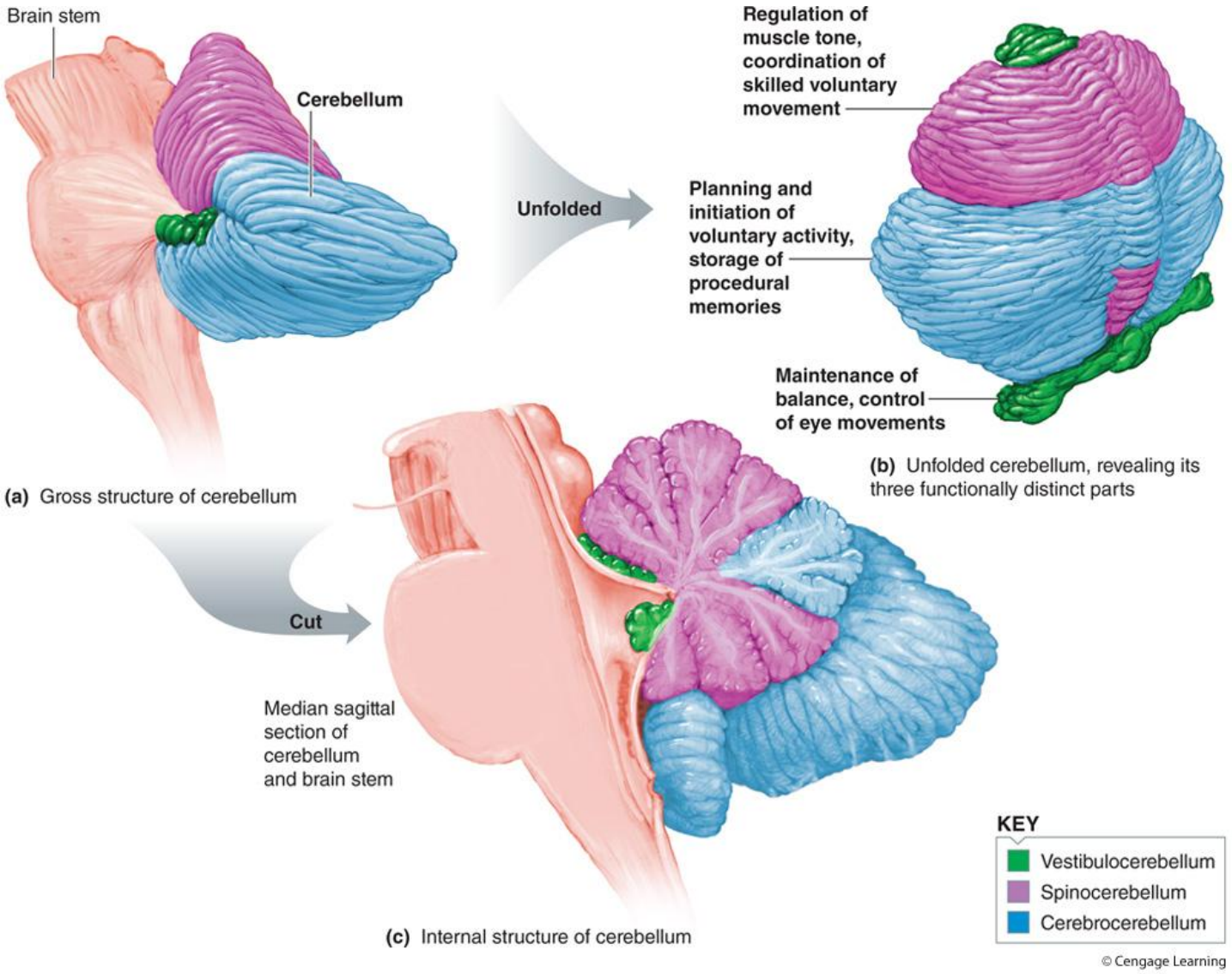
5.8 Cerebellum (소뇌)

Cerebellum



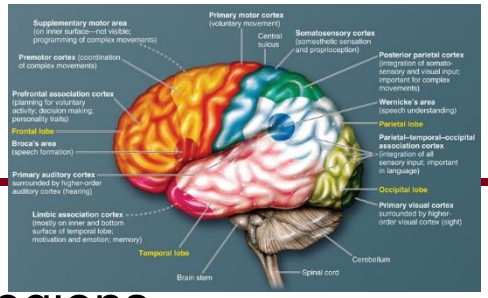
- Highly folded, baseball-sized part lying underneath the cortex's occipital lobe and attached to the back of upper portion of brain stem
- Cerebellum is important for balance, planning, and executing voluntary movement
- Most brain neurons (50-70%) are found in cerebellum
- Three different parts with different roles associated with subconscious (잠재적) control of motor activity
  - Vestibulocerebellum (안뜰소뇌): balance and eye movement
  - Spinocerebellum (척수소뇌): enhances muscle tone and coordinates skilled, voluntary movement
  - Cerebrocerebellum (대뇌소뇌): planning and initiating voluntary activity

Cerebellum


Brain stem
Cerebellum
Unfolded
Regulation of muscle tone, coordination of skilled voluntary movement
Planning and initiation of voluntary activity, storage of procedural memories
Maintenance of balance, control of eye movements
(a) Gross structure of cerebellum
(b) Unfolded cerebellum, revealing its three functionally distinct parts
Cut
Median sagittal section of cerebellum and brain stem
(c) Internal structure of cerebellum
KEY
Vestibulocerebellum
Spinocerebellum
Cerebrocerebellum
© Cengage Learning

5.9 Brain Stem (뇌간)

Brain stem



- Vital link between spinal cord and higher brain regions
- Consists of midbrain, pons, and medulla
- Five functions of brain stem

1) Area where most cranial nerve (뇌신경) arise
   - Cranial nerves supply structures in head and neck with both sensory and motor fibers
   - Cranial nerve X (vagus nerve): supply chest and belly cavities (PNS)

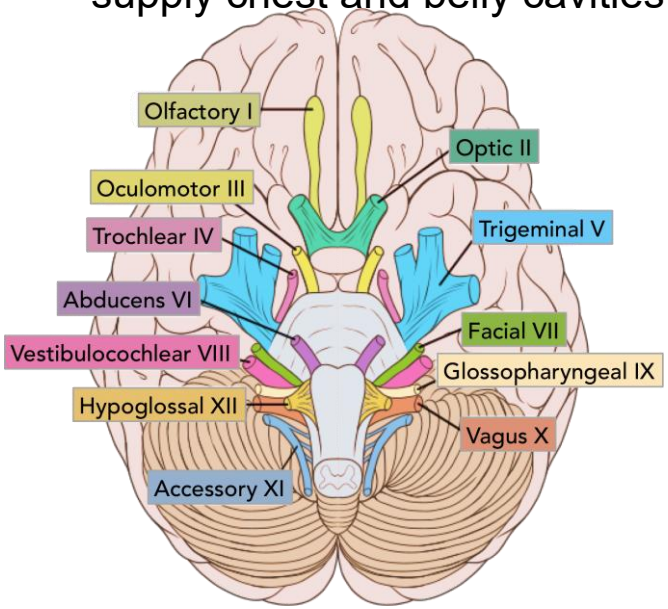
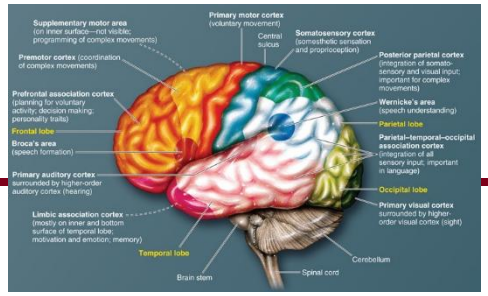


TABLE 5-3 Functions of Cranial Nerves

Number	Name	Fiber Types	Functions
I	Olfactory	Sensory	■ Smell
II	Optic	Sensory	■ Vision
III	Oculomotor	Mixed (mainly motor)	● Eyeball and eyelid movement, pupil constriction, change of lens shape for near vision.
			■ Proprioception (awareness of position of body parts)
IV	Trochlear	Mixed (mainly motor)	● Eyeball movement
			■ Proprioception
V	Trigeminal	Mixed	● Chewing
			■ Somatic sensations (touch, pressure, pain, and temperature) of face and mouth
VI	Abducens	Mixed (mainly motor)	● Eyeball movement
			■ Proprioception
VII	Facial	Mixed	● Facial expression, secretion of saliva and tears
			■ Taste from front of tongue
VIII	Vestibulocochlear	Sensory	■ Hearing, sense of equilibrium
IX	Glossopharyngeal	Mixed	● Swallowing, secretion of saliva
			■ Taste from back of tongue, somatic sensation of oral cavity, blood-pressure monitoring
X	Vagus	Mixed	● Efferent output for skeletal muscles of pharynx (throat) and larynx (voice box) and for smooth muscle and glands of thoracic and abdominal organs and for cardiac muscle of heart
			■ Afferent input from thoracic and abdominal organs, blood-pressure monitoring
XI	Accessory	Motor	● Efferent output for skeletal muscles of pharynx, larynx, neck, and shoulder
XII	Hypoglossal	Motor	● Tongue movement

■ Carried by afferent fibers ● Carried by efferent fibers

Brain stem



- Five functions of brain stem

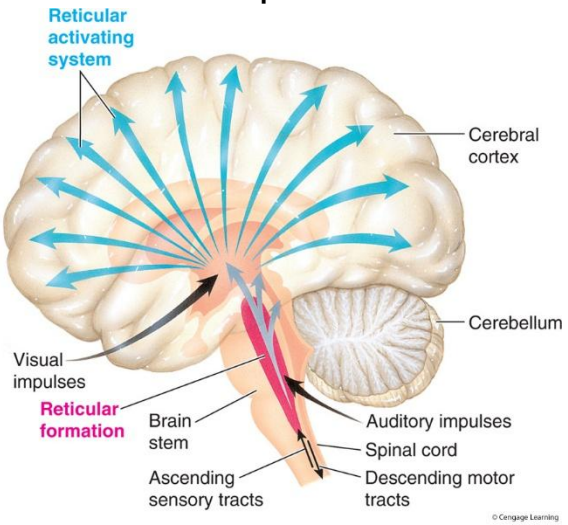
2) Contains neuronal *centers* that control heart and blood vessel function, respiration, and digestive activities (mainly medulla)

3) Helps regulating muscle reflexes involved in equilibrium and posture (mainly midbrain)

4) *Reticular formation* (망상체), widespread network of interconnected neurons, runs throughout the brain stem and into the thalamus

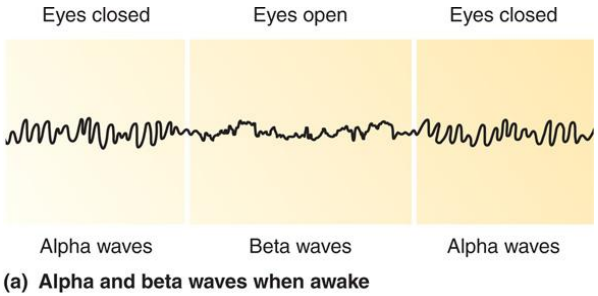
- central hub for arousal, motor control, and autonomic functions
- functionally divided into ascending reticular activating system (ARAS) (maintains cortical arousal, consciousness, and wakefulness) and descending reticular system (DRS) (modulates motor activity, posture, and autonomic responses
- ARAS: midbrain + upper pons
- DRS: pons, medulla

5) Contains centers that govern sleep within brain stem and hypothalamus (mainly pons)

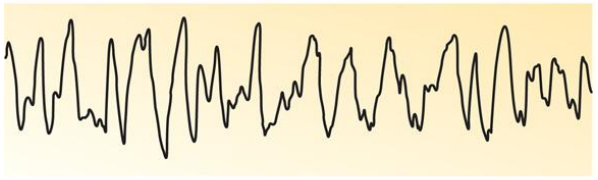


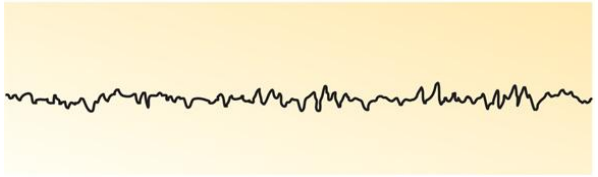
Electroencephalogram (EEG), arousal, and brain stem

- EEG is a record of postsynaptic activity in cortical neurons
- Extracellular current flow from electrical activity within the cerebral cortex can be detected



(a) Alpha and beta waves when awake


(b) Delta waves when in slow-wave sleep, stage 4


(c) Waves during paradoxical sleep

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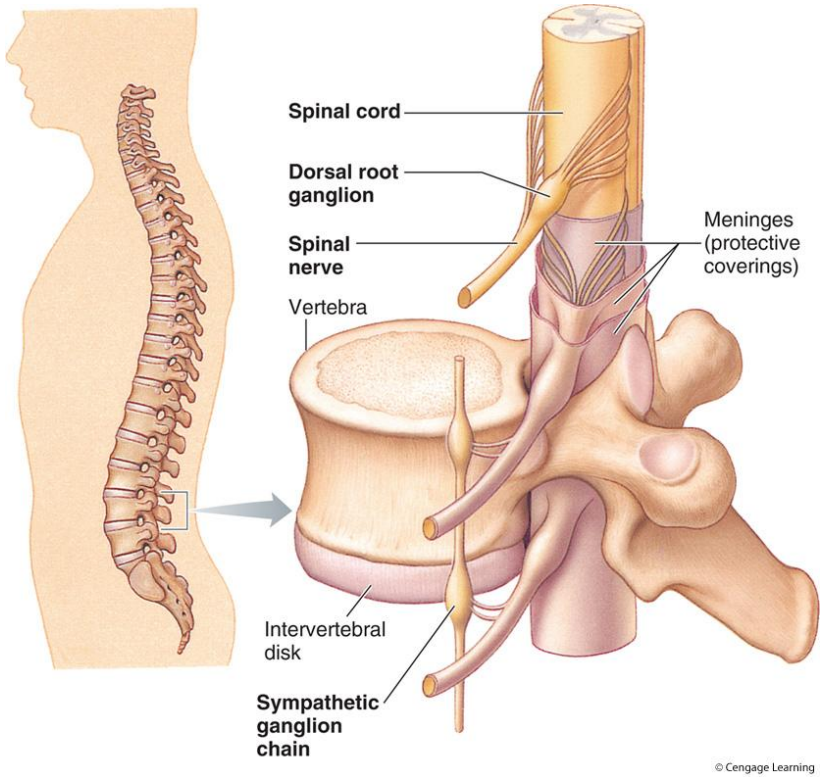
- Sleep is an active process consisting of alternating periods of slow-wave (서파수면, 숙면) and paradoxical sleep (렘수면)
- Sleep-wake cycle is controlled by interaction between three neural systems: arousal system in reticular formation, slow-wave sleep center (hypothalamus), and paradoxical sleep center (pons)

5.10 Spinal Cord (척수)

Spinal cord

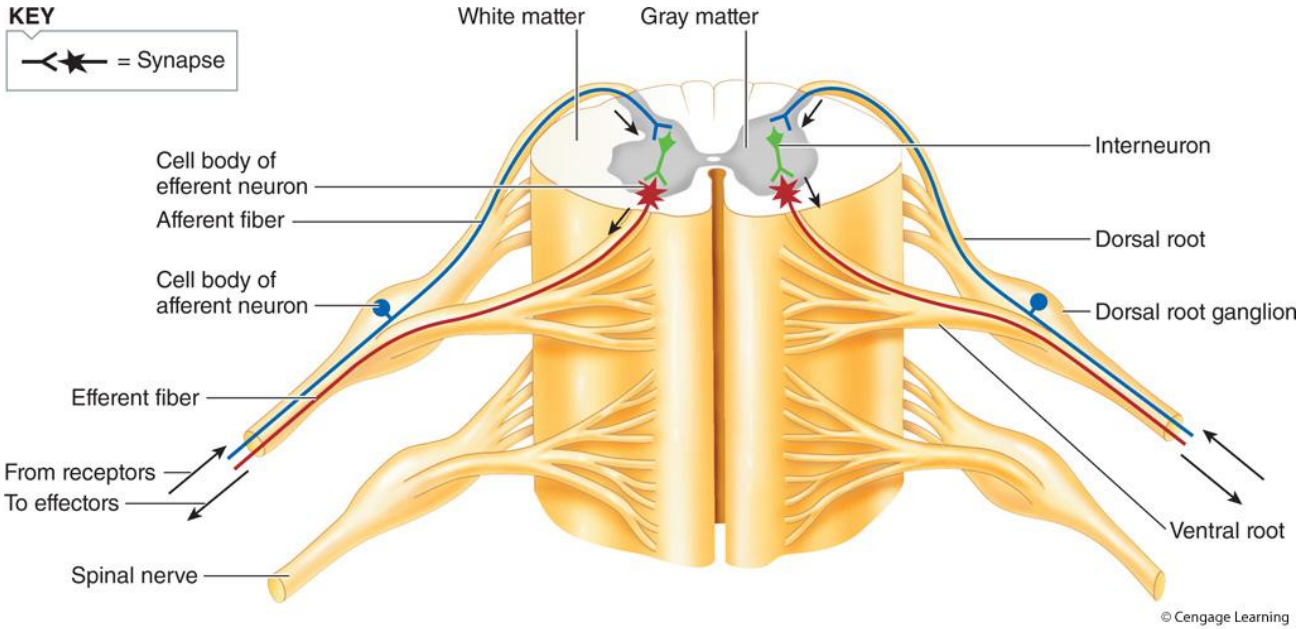
- Long, slender cylinder of nerve tissue extending from the brain stem (45 cm long and 1 to 1.5 cm wide)
- Extends through the vertebral canal (척주관) and is connected to the spinal nerves of PNS

- Vertebra: Bone protecting the spinal cord
- IVD: Cushion between vertebrae
- Meninges: Protective membranes of CNS
- Dorsal root ganglion: Sensory neuron cell body cluster
- Sympathetic ganglion chain: Autonomic ganglia for fight-or-flight

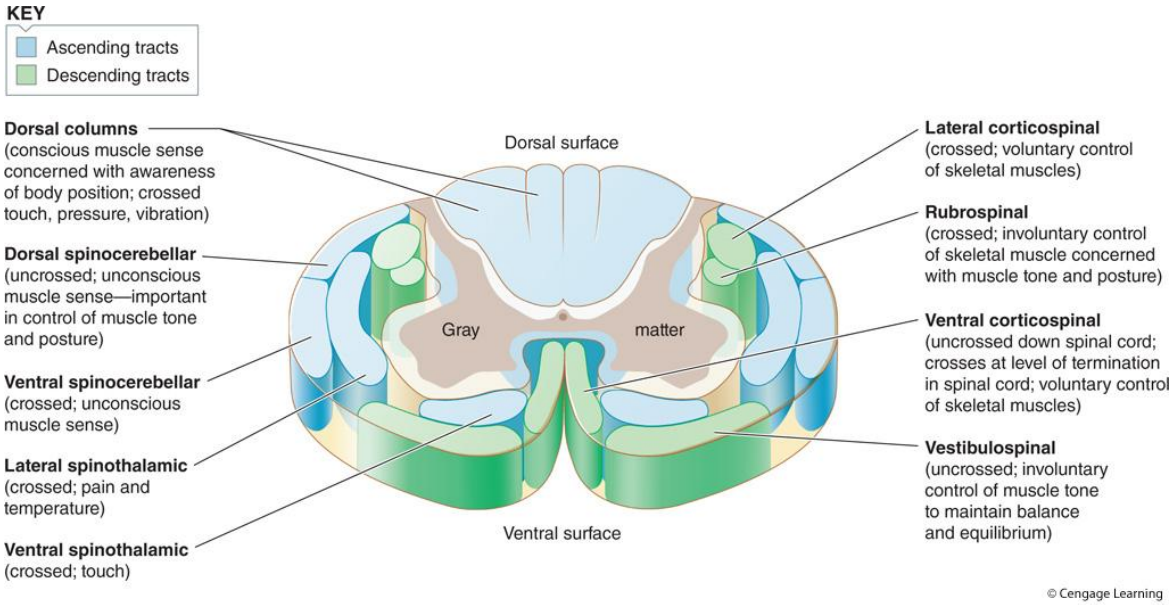


Spinal cord anatomy

- Cross-sectional anatomy is generally the same throughout its length
- White matter forms outer shell capping a butterfly-shaped, inner gray matter
- Gray matter: neuronal cell bodies, their dendrites, and glial cells
- White matter is organized into tracts, bundles of nerve fibers with similar function
  - Ascending / descending / ventral spinocerebellar / ventral corticospinal tracts

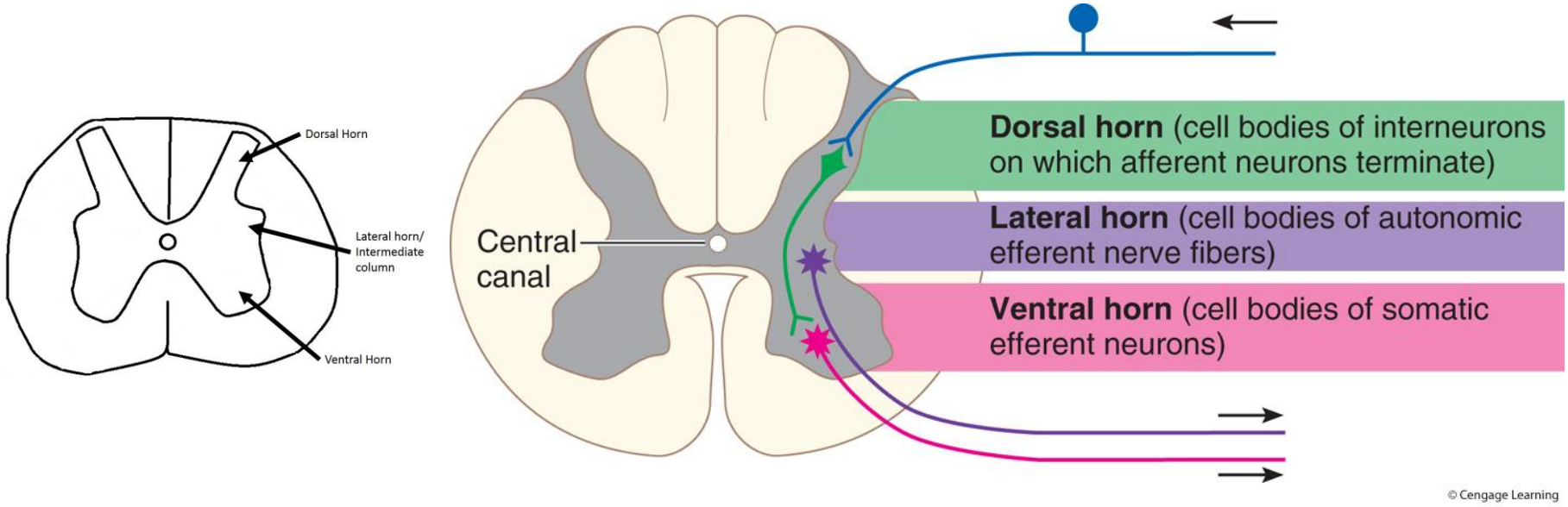


Spinal cord anatomy – white matter



- Ascending: cord to brain, transmit signals from afferent input to brain
- Descending: brain to cord, relay messages from brain to efferent neurons
- Ventral spinocerebellar: originating from spinal cord, runs up the ventral (front body) side of the cord, and ends at cerebellum
- Ventral corticospinal tract: originating from motor region of cerebral cortex, runs down the ventral side of spinal cord, and ends in spinal cords connected to skeletal muscles

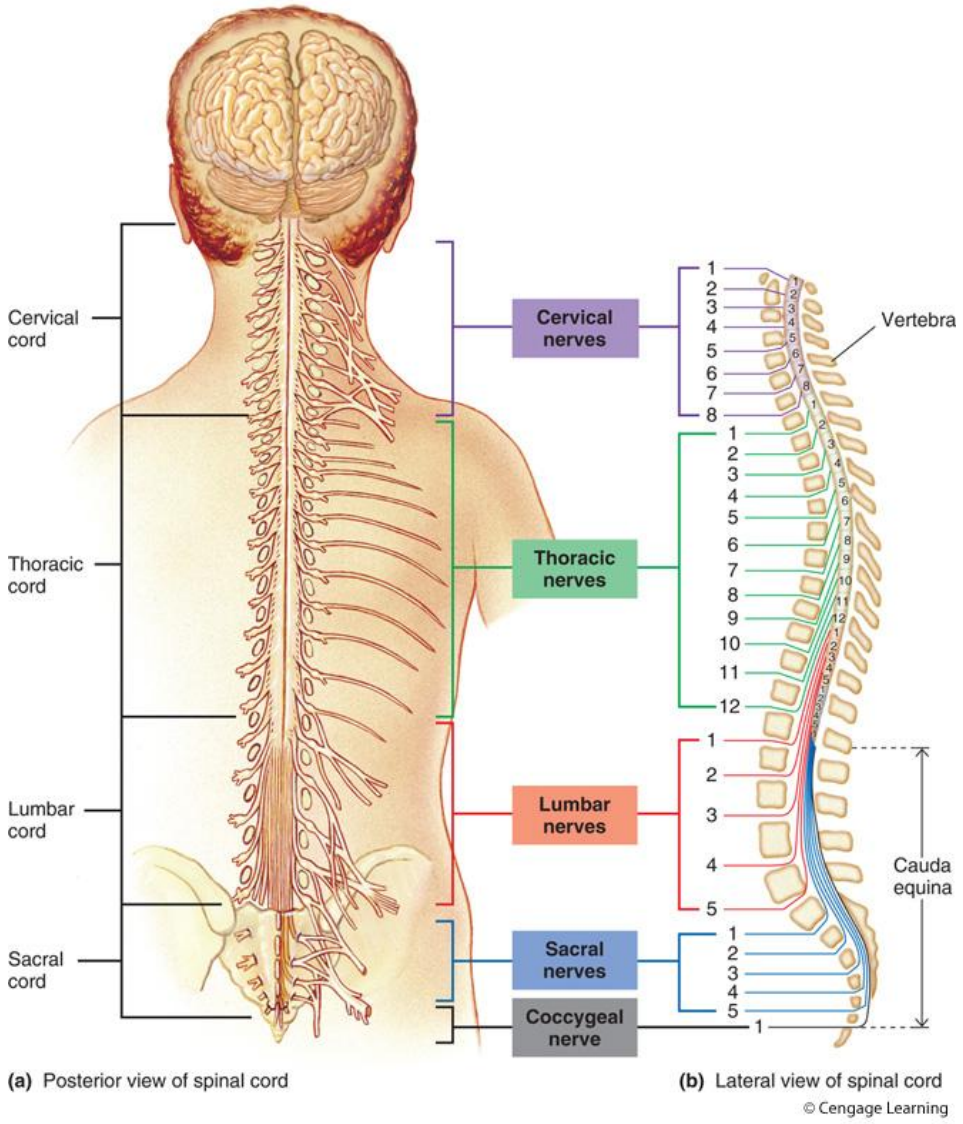
Spinal cord anatomy – grey matter



- Central canal: filled with CSF
- Dorsal (posterior) horn: contains cell bodies of interneurons on which afferent neurons terminate
- Ventral (anterior) horn: contains cell bodies of the efferent neurons supplying skeletal muscles
- Lateral horn: contains cell bodies of autonomic efferent nerve fibers supplying cardiac and smooth muscle, and exocrine glands

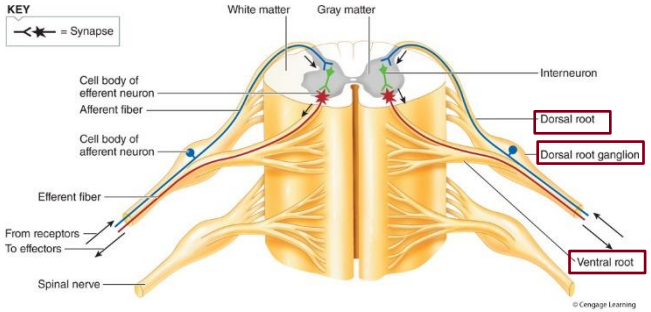
Spinal nerves (in PNS)

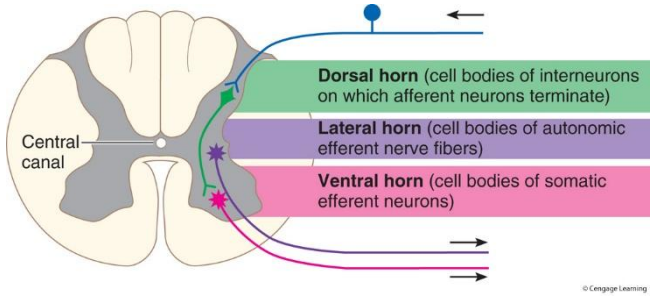
- 8 pairs of cervical (neck) nerves, 12 thoracic (chest) nerves, 5 lumbar (abdominal) nerves, 5 sacral (pelvic) nerves, and 1 coccygeal (tailbone) nerve
- Vertebral column grows 25 cm longer than spinal cord. Therefore, most of spinal nerve roots descend along the cord, lower than spinal cord.
- Spinal tap (lumbar puncture): L3-L4 or L4-L5, lower than L2 not to hurt the spinal cord



Spinal nerves (in PNS)

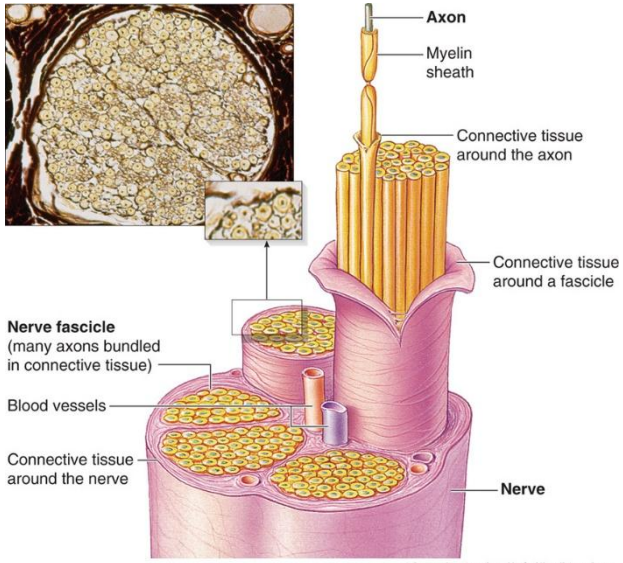
- Spinal nerves carry both afferent and efferent fibers
  - Spinal nerves connect each side of the spinal cord by dorsal root and ventral root
- Afferent fibers from peripheral receptors enter the spinal cord through dorsal root
- The clustered cell bodies for the afferent neurons are in dorsal root ganglion
  - Ganglion: a collection of neuronal cell bodies outside CNS
  - Nucleus or Center: a collection of neuronal cell bodies within CNS
- The cell bodies for the efferent neurons locate in gray matter and their efferent fibers to muscles and glands exit through ventral root

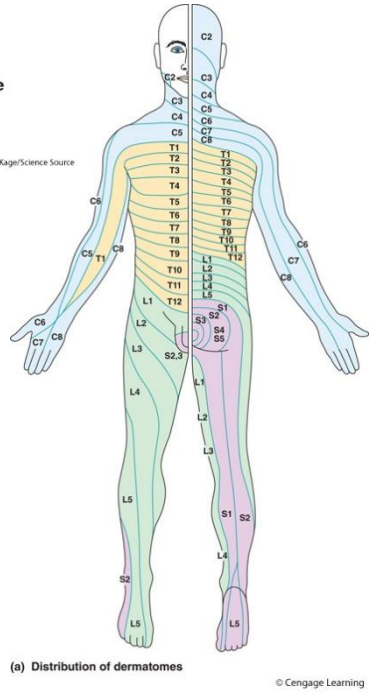




Structure of a nerve

- Nerve
  - A bundle of axons of peripheral neurons, enclosed by connective tissue
  - A nerve does not contain a complete neuron, only the axonal portions of multiple neurons
  - There is no nerve in CNS
- Nerve fascicle (신경다발)
  - Bundle of nerve fibers (axons) within PNS, wrapped in a protective connective tissue sheath
- 31 pairs of spinal nerves and 12 pairs of cranial nerves arising from the brain stem constitute the PNS
- Dermatome
  - specific area of skin that receives sensory input from a single spinal nerve root





Structure of the spinal cord – just for summary



https://www.youtube.com/watch?v=fbt3H3JxRMA

Spinal cord and innate reflexes (선천 반사)

- The spinal cord is responsible for integration of many innate reflexes
  - Reflex: any response occurring automatically without conscious effort
  - Spinal cord (1) serves as a link for transmission of information between brain and rest of the body and (2) integrates reflex activity between afferent input and efferent output without involving the brain
  - Reflex arc: the neural pathway involved in reflex activity
    - Five components of basic principles of reflex
    - Sensory receptor: responds to a stimulus and produce AP
    - Afferent pathway: action potential is relayed
    - Integrating center: processes all inputs and make a decision about response
    - Efferent pathway: instruction signals are transmitted
    - Effector organ: muscle or gland, carrying out a response

Categories of reflexes

- Spinal or cranial
  - Spinal reflexes: reflex is integrated by the spinal cord (knee-jerk (patellar) reflex, withdrawing a hand from a hot stove, or pulling a foot away from a sharp object)
  - Cranial reflex: reflex is subconsciously integrated by the brain (the pupil reflex (constricting in bright light), the corneal reflex (blinking when the eye is touched), sneezing, coughing, and gagging)
- Innate or conditioned
  - Innate reflex: unlearned (withdrawal reflex, pupil reflex)
  - Conditioned reflex: result of learning (saliva secretion after smelling food, sports skills)
- Somatic or autonomic
  - Somatic reflex: Efferent neuron is skeletal muscle
  - Autonomic reflex: autonomic nervous system to smooth muscle, cardiac muscle, or glands
- Monosynaptic or polysynaptic
  - Monosynaptic: integration center is the single synapse in spinal cord (patellar reflex, ankle-jerk reflex, biceps reflex)
  - Polysynaptic: involves at least one interneuron (any other than mono-)